Code Generation

1

Code Generation at a Glance

- At this point, we have optimized IR code that needs to be converted into the target language (e.g. assembly, machine code).
- Goal of this stage:
  - Choose the appropriate machine instructions for each IR instruction.
  - Allocate finite machine resources (registers, caches, etc.)
  - Implement low-level details of the runtime environment.
  -
- Machine-specific optimizations are often done here, though some are treated as part of a final optimization phase.

2

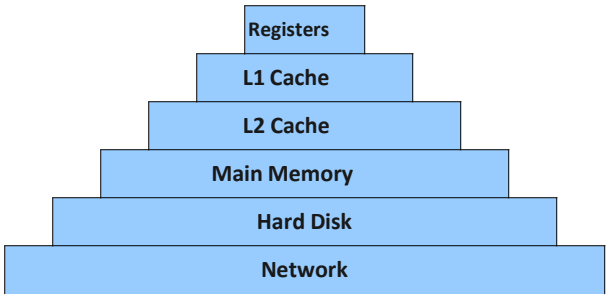
Overview

- **Register Allocation**
  - How to assign variables to finitely many registers? What to do when it can't be done? How
  - to do so efficienty?
  -
- **Garbage Collection**
  - How to detect reclaimable memory? How to
  - reclaim memory efficiently?

3

The Memory Hierarchy

- **Idea**: Try to get the best of all worlds by using multiple types of memory.



256B - 8KB	0.25 – 1ns
16KB – 64KB	1ns – 5ns
1MB - 4MB	5ns – 25ns
4GB – 256GB	25ns – 100ns
500GB+	3 – 10ms
HUGE	10 – 2000ms

4

Registers

- Most machines have a set of **registers**, dedicated memory locations that
  - can be accessed quickly,
  - can have computations performed on them, and exist in
  - small quantity.
- Using registers intelligently is a critical step in any compiler.
  - A good register allocator can generate code orders of magnitude better than a bad register allocator.

5

Register Allocation

- There are an unlimited number of variables.
- On a physical machine there are a small number of registers:
  - x86 has four general-purpose registers and a number of specialized registers.
  - MIPS has twenty-four general-purpose registers and eight special-purpose registers.
- **Register allocation** is the process of assigning variables to registers and managing data transfer in and out of registers.

6

Challenges in Register Allocation

- **Registers are scarce**.
  - Often substantially more IR variables than registers. Need to find a way
  - to reuse registers whenever possible.
- **Registers are complicated**.
  - x86: Each register made of several smaller registers; can't use a register and its constituent registers at the same time.
  - x86: Certain instructions must store their results in specific registers; can't store values there if you want to use those instructions.
  - MIPS: Some registers reserved for the assembler or operating system.
  - Most architectures: Some registers must be preserved across function calls.
  -

7

An Initial Register Allocator

- **Idea**: Store every value in main memory, loading values only when they're needed.
- To generate a code that performs a computation:
  - Generate **load** instructions to pull the values from main memory into registers.
  - Generate code to perform the computation on the registers.
  - Generate **store** instructions to store the result back into main memory.

8

Building a Better Allocator

- **Goal**: Try to hold as many variables in registers as possible.
  - Reduces memory reads/writes.
  - Reduces total memory usage.
- We will need to address these questions:
  - Which registers do we put variables in? What do we do
  - when we run out of registers?

9

Register Consistency

- At each program point, each variable must be in the same location.
  - Does **not** mean that each variable is always stored in the same location!
- At each program point, each register holds at most one live variable.
  - Can assign several variables the same register if no two of them ever will be read together.

10

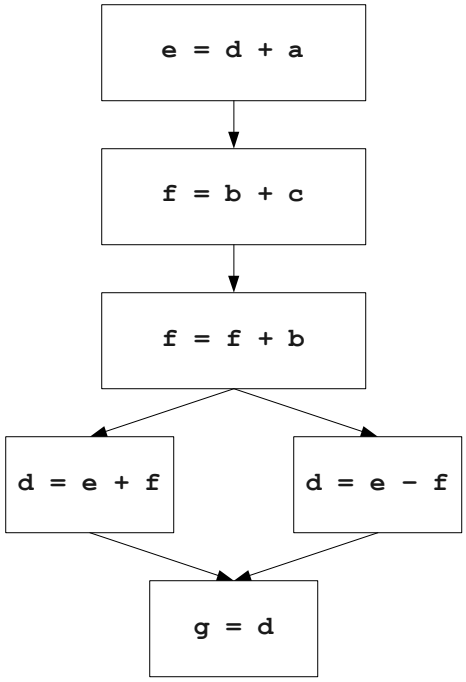
Live Ranges and Live Intervals

- Recall: A variable is **live** at a particular program point if its value may be read later before it is written.
  - Can find this using global liveness analysis.
- The **live range** for a variable is the set of program points at which that variable is live.
- The **live interval** for a variable is the smallest subrange of the IR code containing all a variable's live ranges.
  - A property of the IR code, **not** the CFG.
  - Less precise than live ranges, but simpler to work with.

11

Live Ranges and Live Intervals

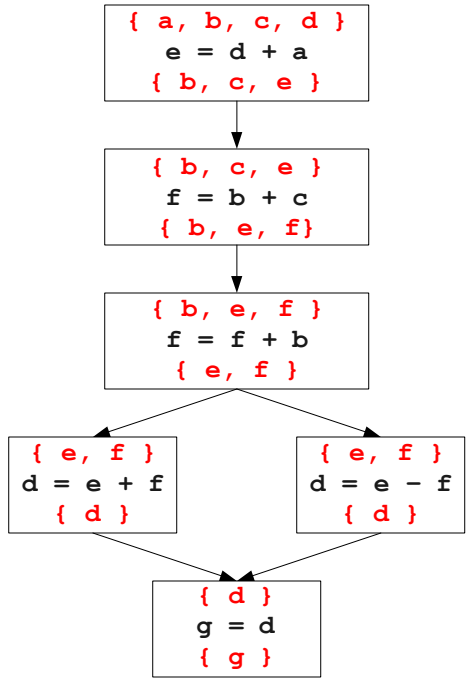
```
    e = d + a
    f = b + c
    f = f + b
    IfZ e Goto _L0
    d = e + f
    Goto _L1;
_L0:
    d = e - f
_L1:
    g = d
```



12

Live Ranges and Live Intervals

```
    e = d + a
    f = b + c
    f = f + b
    IfZ e Goto _L0
    d = e + f
    Goto _L1;
_L0:
    d = e - f
_L1:
    g = d
```



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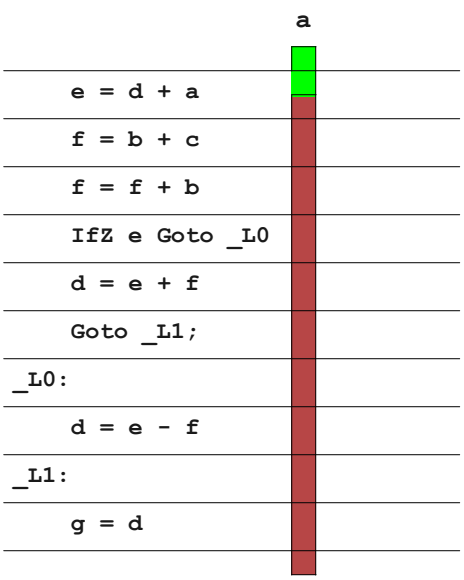
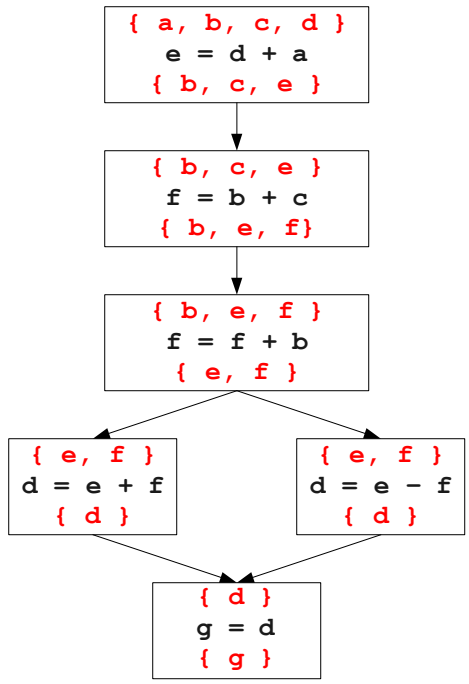
Live Ranges and Live Intervals





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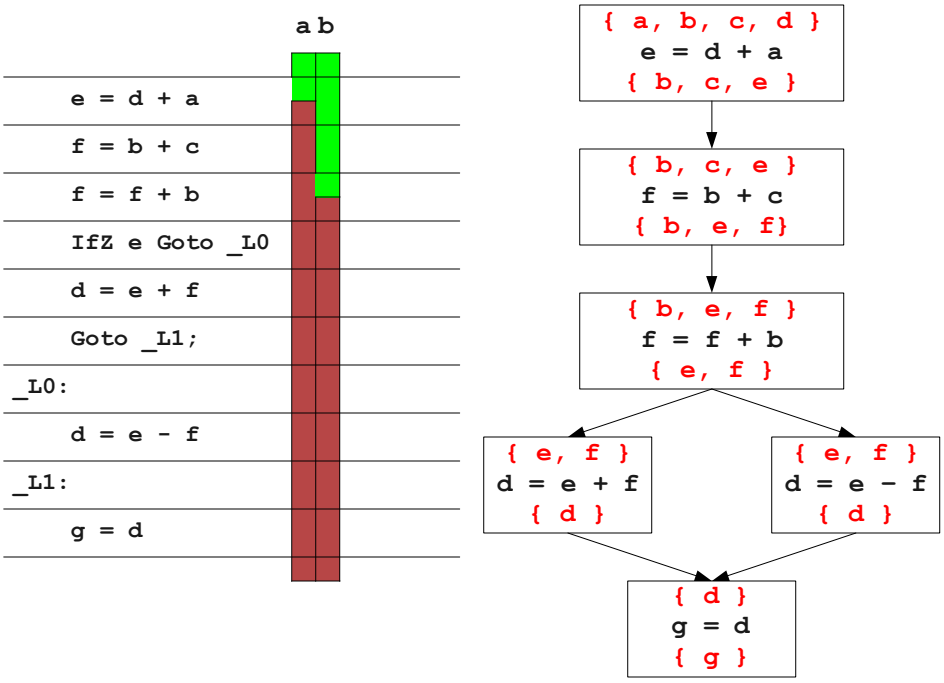
Live Ranges and Live Intervals



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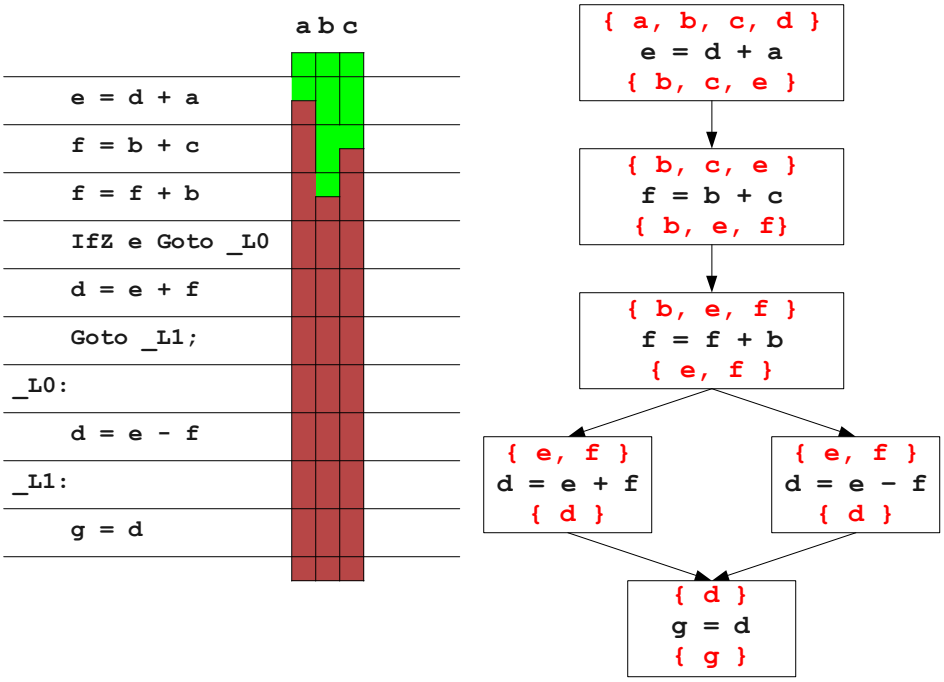
Live Ranges and Live Intervals



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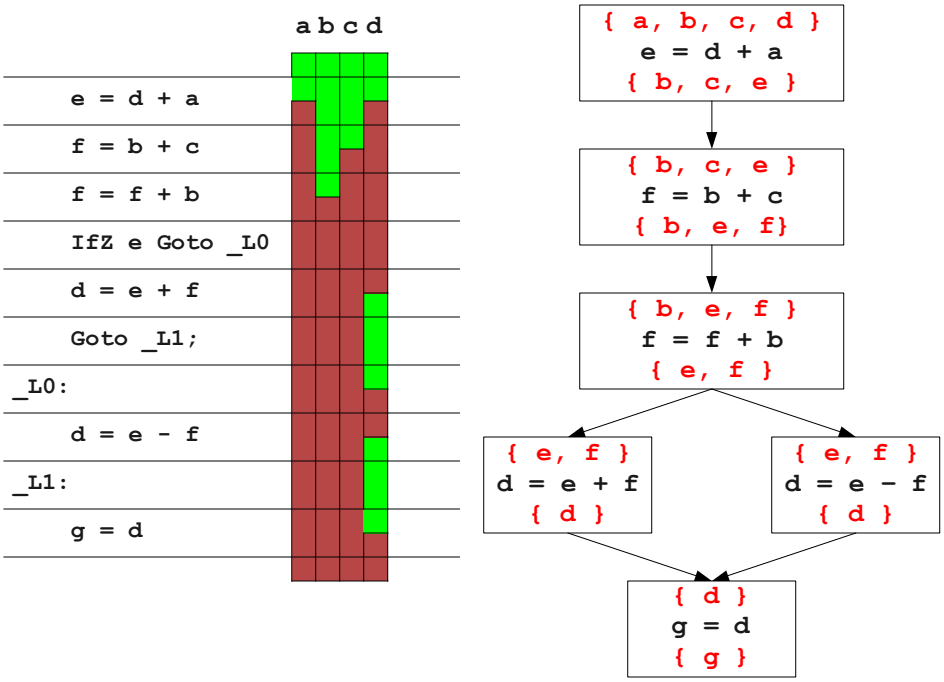
Live Ranges and Live Intervals



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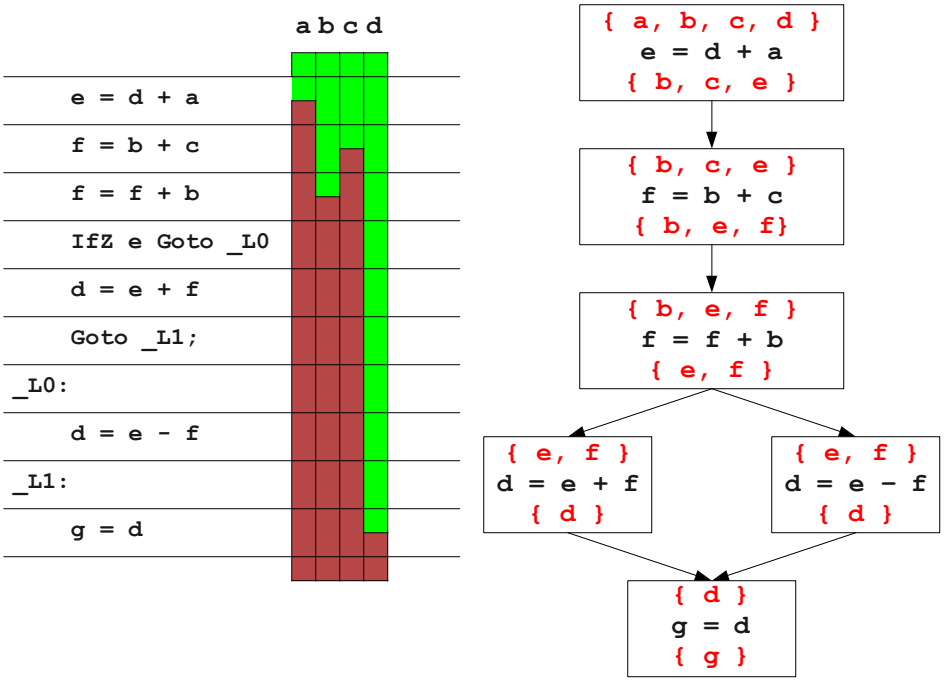
Live Ranges and Live Intervals



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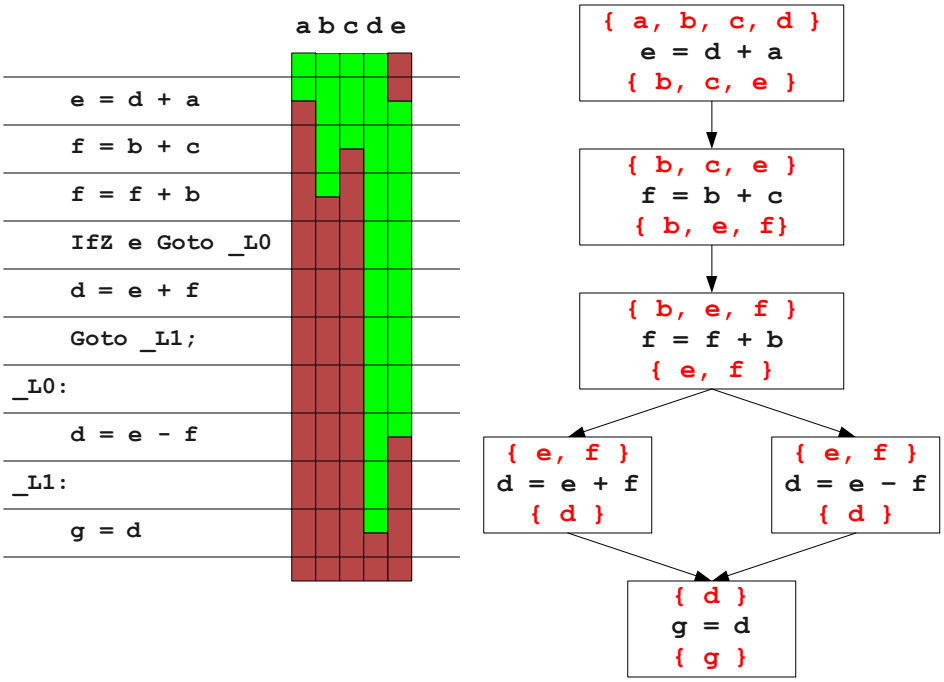
Live Ranges and Live Intervals



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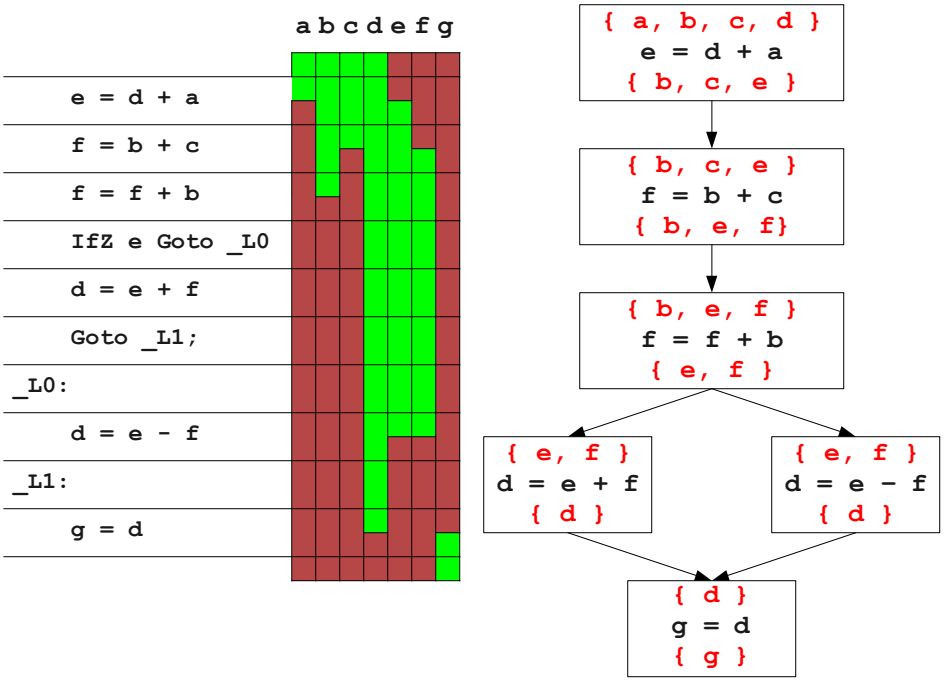
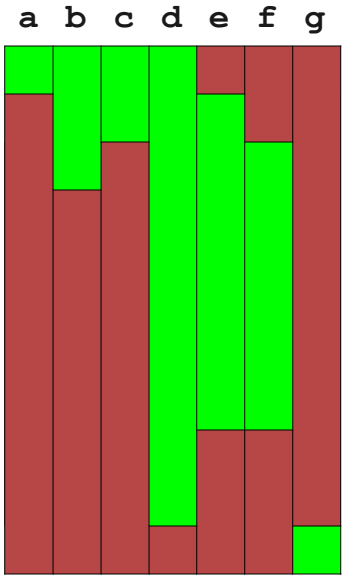
Live Ranges and Live Intervals



20

Register Allocation with Live Intervals

- Given the live intervals for all the variables in the program, we can allocate registers using a simple greedy algorithm.
- Idea: Track which registers are free at each point.
- When a live interval begins, give that variable a free register.
- When a live interval ends, the register is once again free.
- We can't always fit everything into a register; we'll see what do to in a minute.



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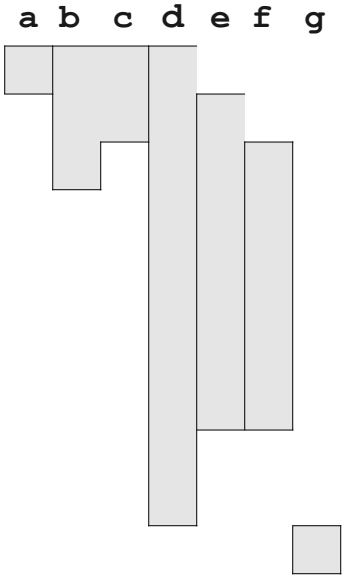
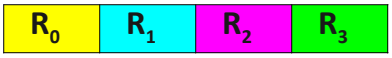
Register Allocation with Live Intervals



Free Registers



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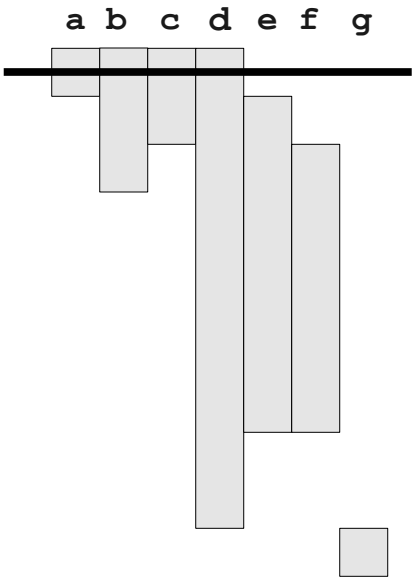
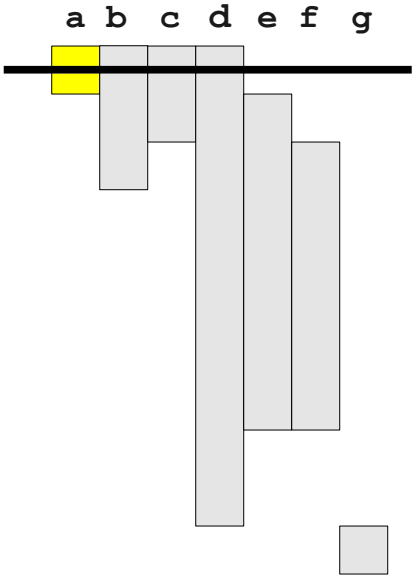
Register Allocation with Live Intervals



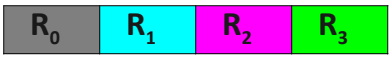
Free Registers

23

Register Allocation with Live Intervals



Free Registers



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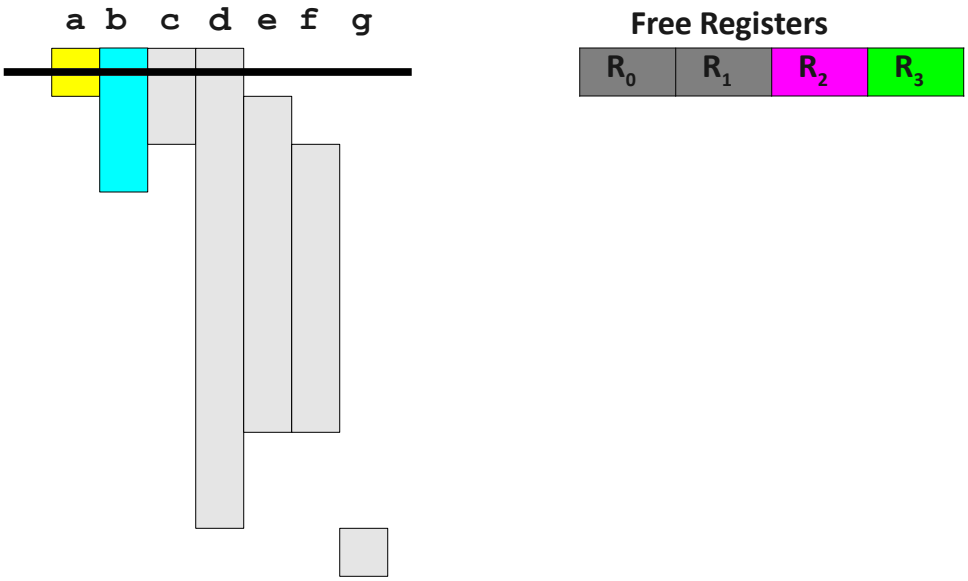
Register Allocation with Live Intervals



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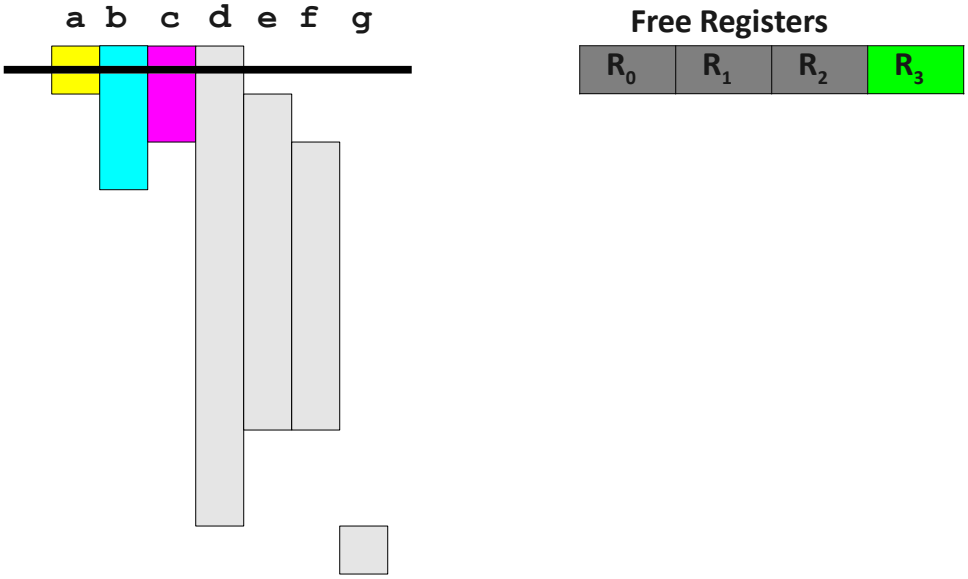

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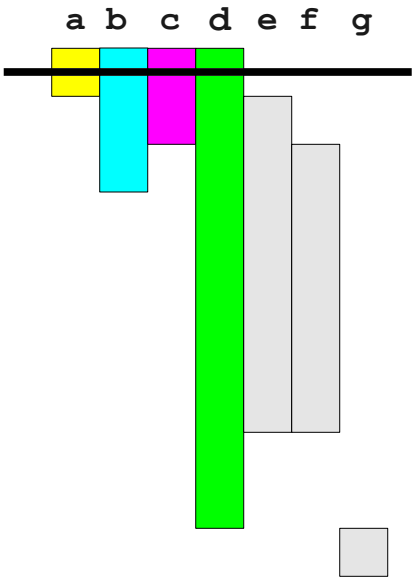
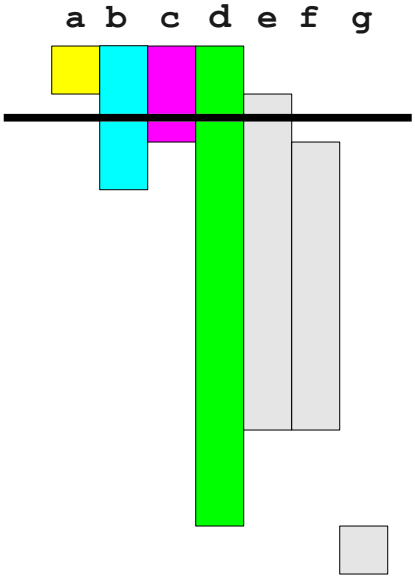


Free Registers

R_0	R_1	R_2	R_2

27

Register Allocation with Live Intervals

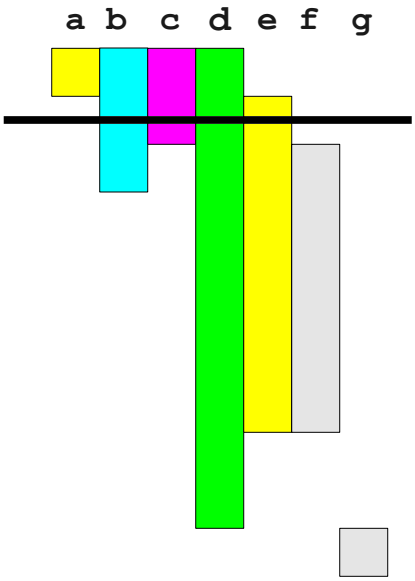


Free Registers

R_0	R_1	R_2	R_2

28

Register Allocation with Live Intervals

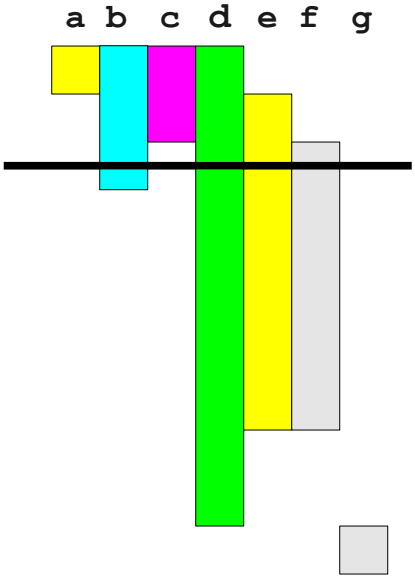


Free Registers

R_0	R_1	R_2	R_2

29

Register Allocation with Live Intervals

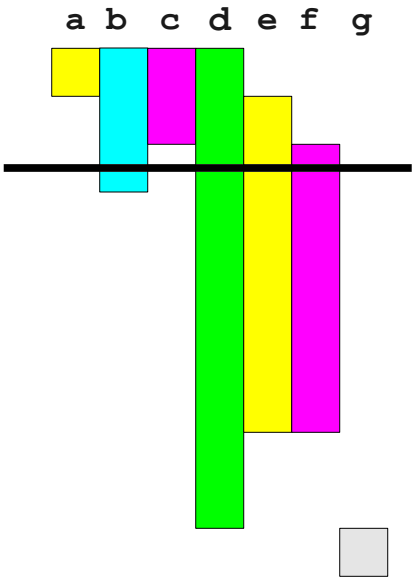


Free Registers

R_0	R_1	R_2	R_2

30

Register Allocation with Live Intervals

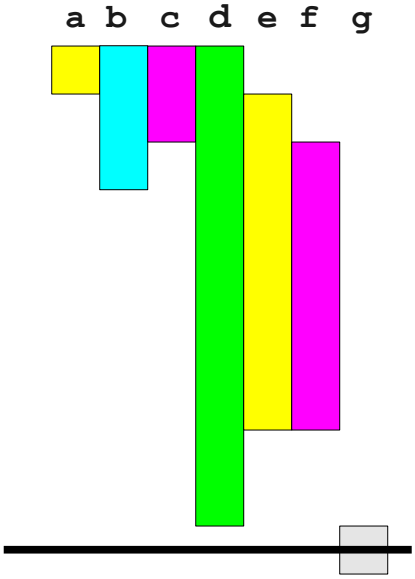


Free Registers

R_0	R_1	R_2	R_2

31

Register Allocation with Live Intervals

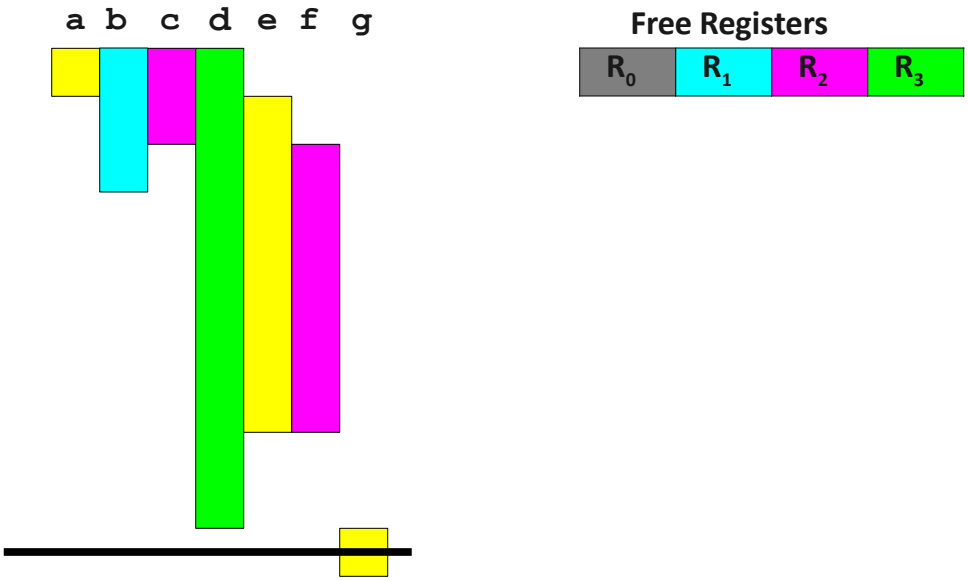


Free Registers

R_0	R_1	R_2	R_3

32

Register Allocation with Live Intervals



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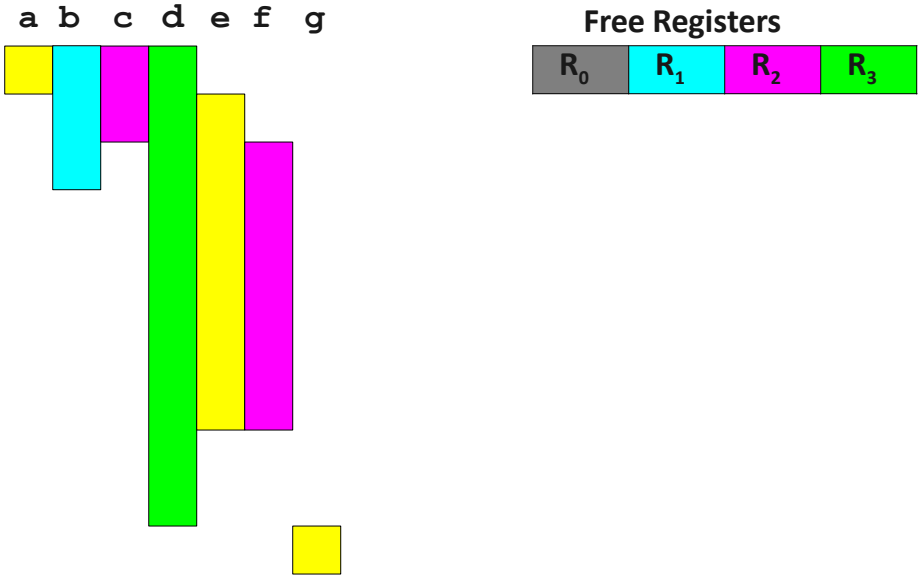
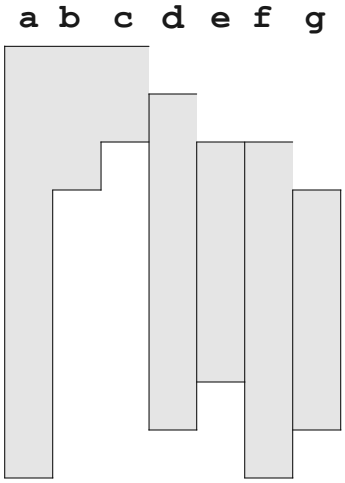
Register Allocation with Live Intervals



34

Another Example

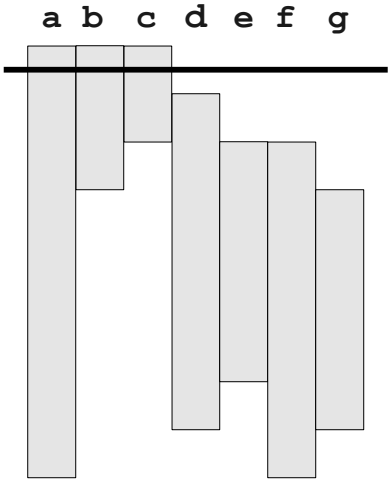


Free Registers

R_0	R_1	R_2

35

Another Example

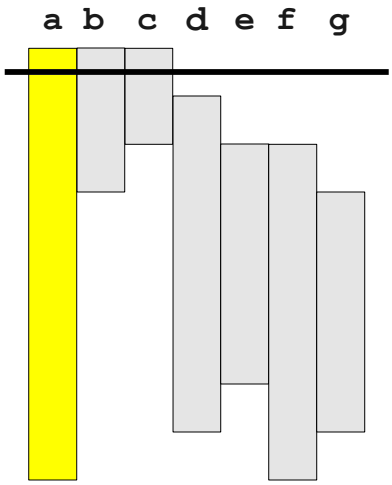


Free Registers

R_0	R_1	R_2

36

Another Example

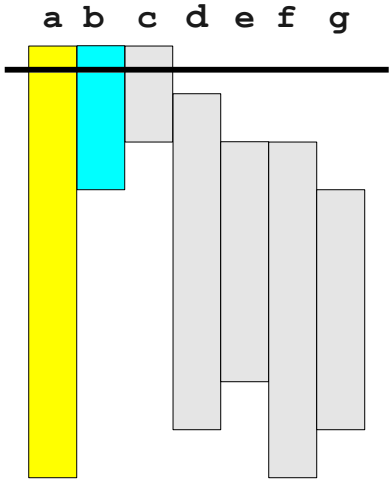


Free Registers

R_0	R_1	R_2

37

Another Example

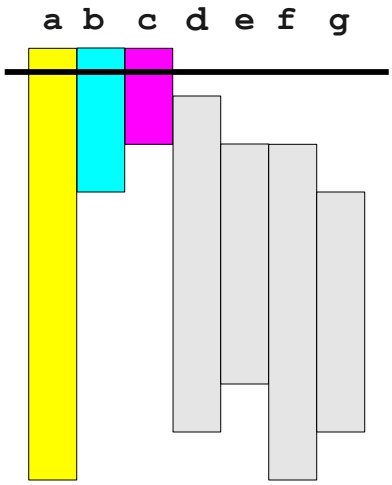


Free Registers

R_0	R_1	R_2

38

Another Example

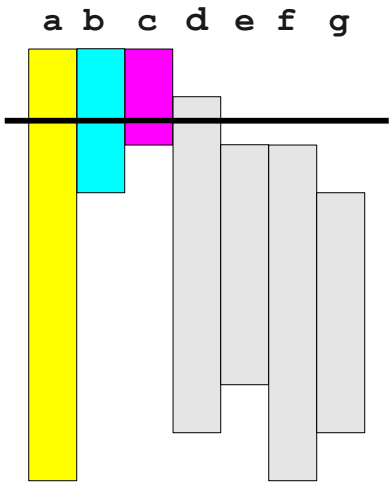


Free Registers

R_0	R_1	R_2

39

Another Example



Free Registers

R_0	R_1	R_2

What do we do now?

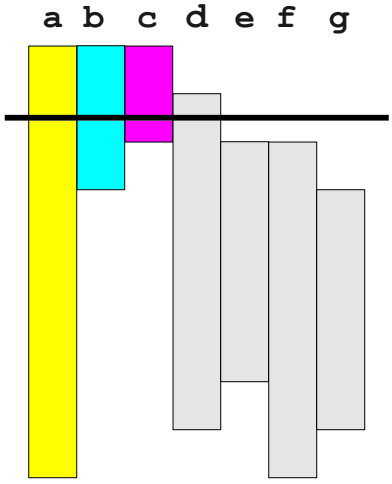
40

Register Spilling

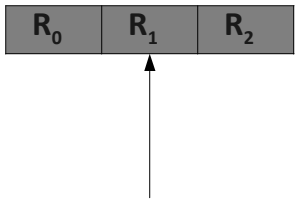
- If a register cannot be found for a variable *v*, we may need to **spill** a variable.
- When a variable is spilled, it is stored in memory rather than a register.
- When we need a register for the spilled variable:
  - Remove some variable from register to memory.
  - Load the variable into the register.
  - When done, write the register back to memory and reload the register with its original value.
- Spilling is slow, but sometimes necessary.

41

Another Example



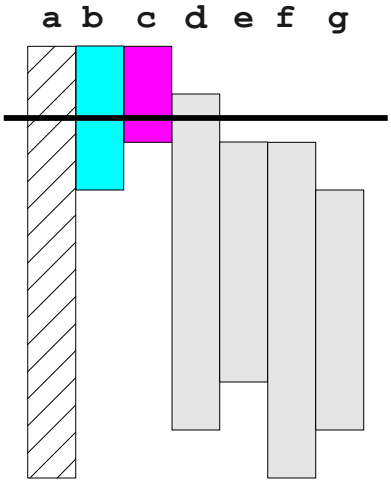
Free Registers



What do we do now?

42

Another Example

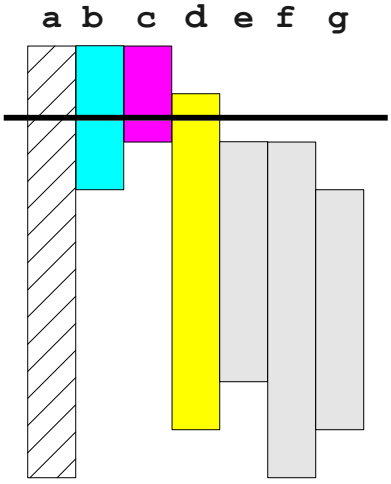


Free Registers

R_0	R_1	R_2

43

Another Example

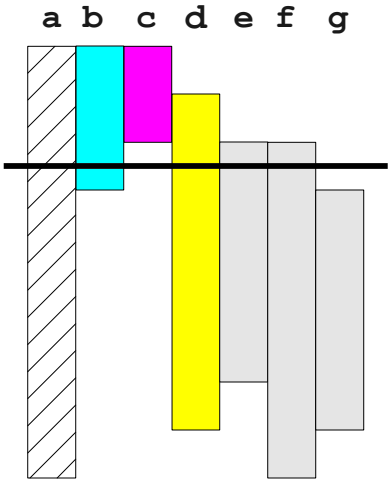


Free Registers

R_0	R_1	R_2

44

Another Example

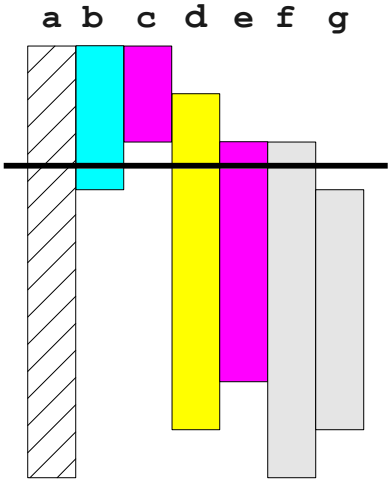


Free Registers

R_0	R_1	R_2

45

Another Example

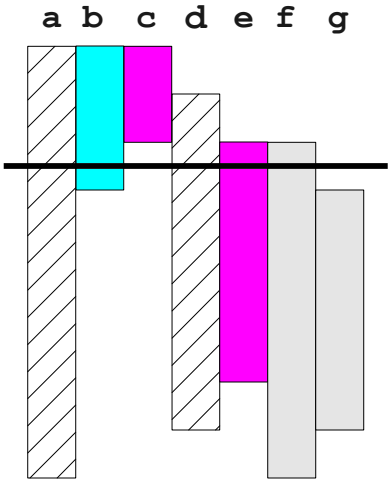


Free Registers

R_0	R_1	R_2

46

Another Example

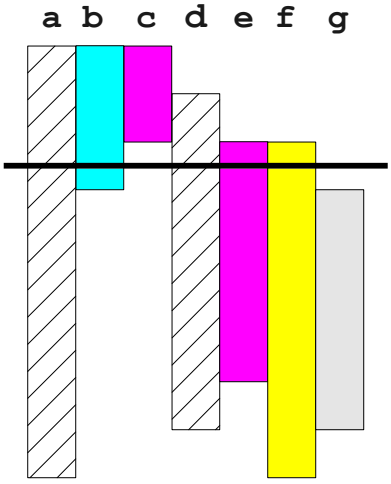


Free Registers

R_0	R_1	R_2

47

Another Example

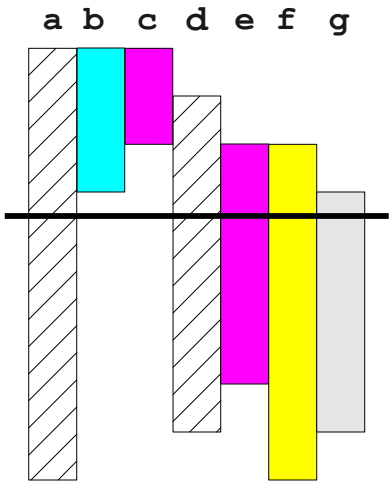


Free Registers

R_0	R_1	R_2

48

Another Example

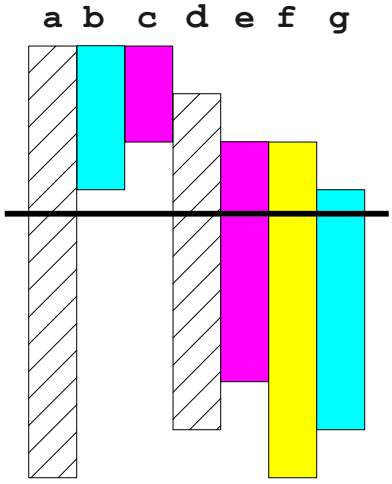


Free Registers

R_0	R_1	R_2

49

Another Example

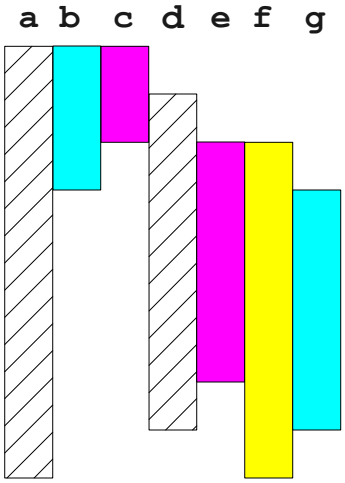


Free Registers

R_0	R_1	R_2

50

Another Example



Free Registers

R_0	R_1	R_2

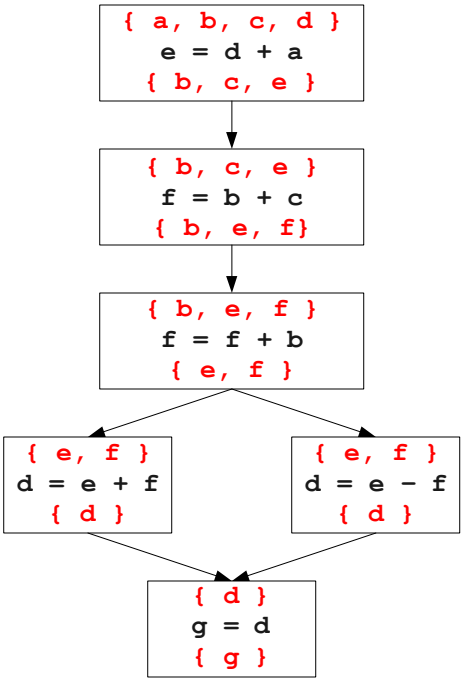
51

Linear Scan Register Allocation

- This algorithm is called **linear scan register allocation** and is a comparatively new algorithm.
- Advantages:
  - Very efficient (after computing live intervals, runs in linear time)
    Produces good code in many instances.
  - Allocation step works in one pass; can generate code during iteration.
  - Often used in JIT compilers like Java HotSpot.
  -
- Disadvantages:
  - Imprecise due to use of live **intervals** rather than live **ranges**. Other
  - techniques known to be superior in many cases.

52

Register Interference Graph



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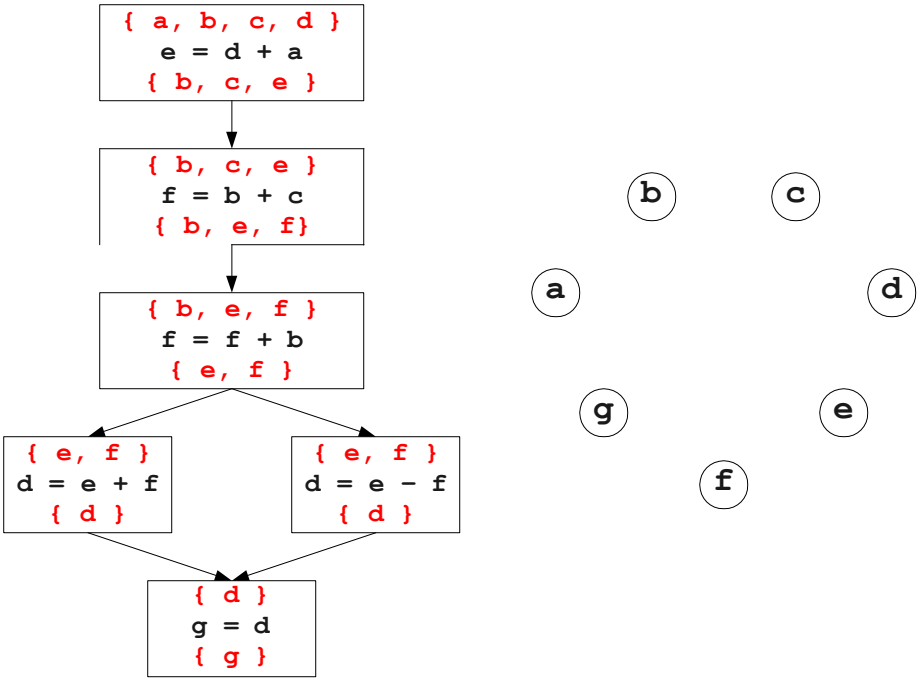

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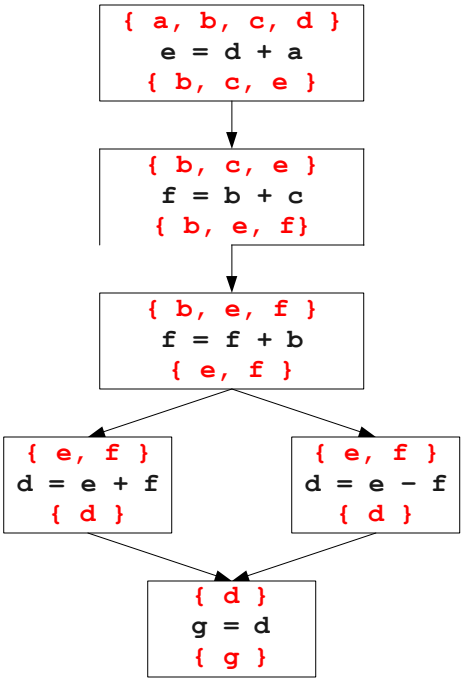

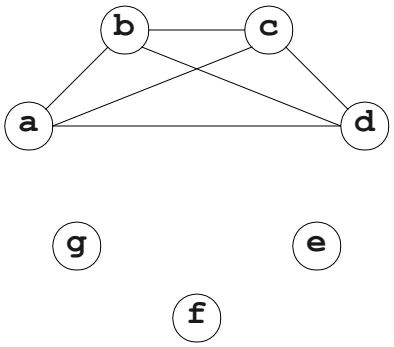

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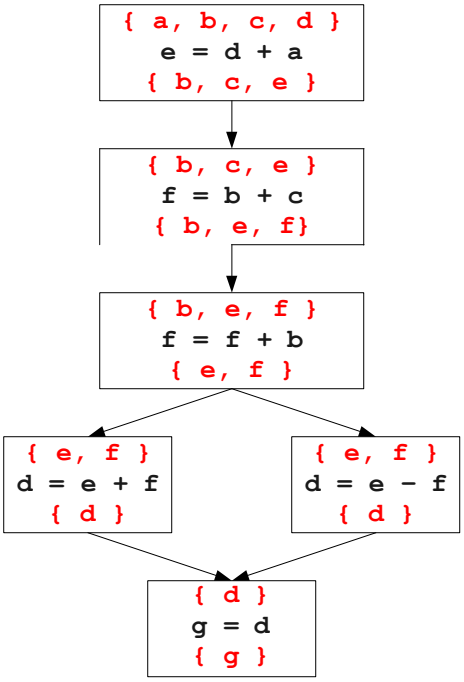

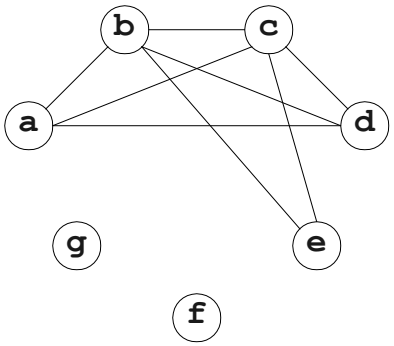

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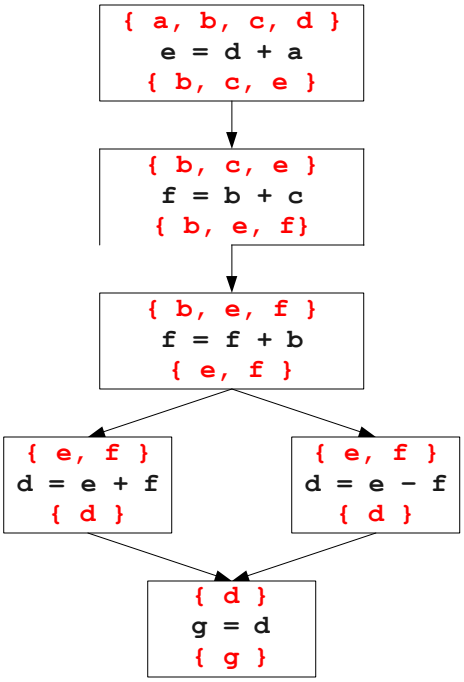

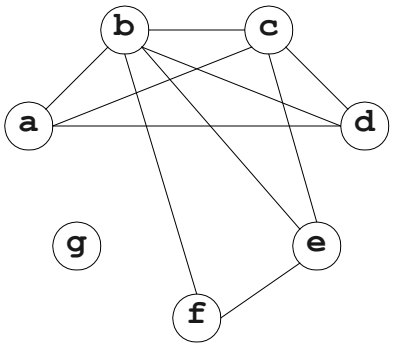

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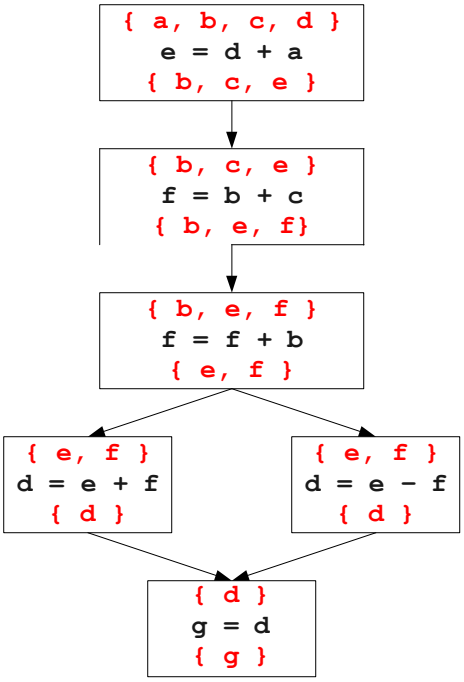

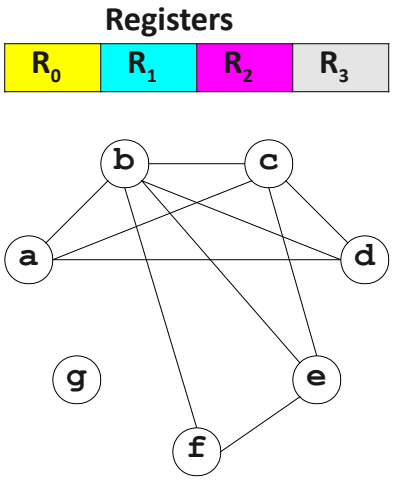

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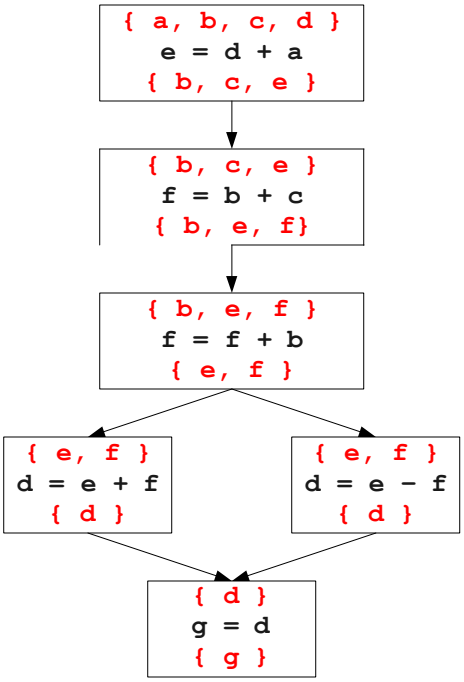
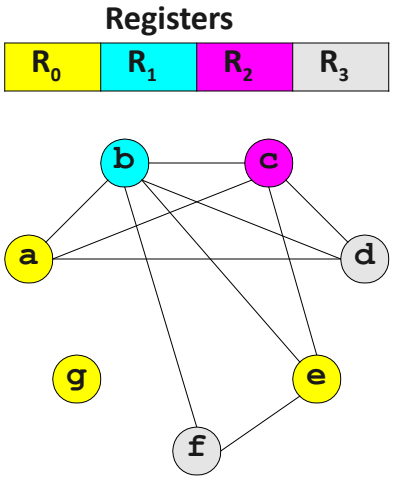
Register Interference Graph





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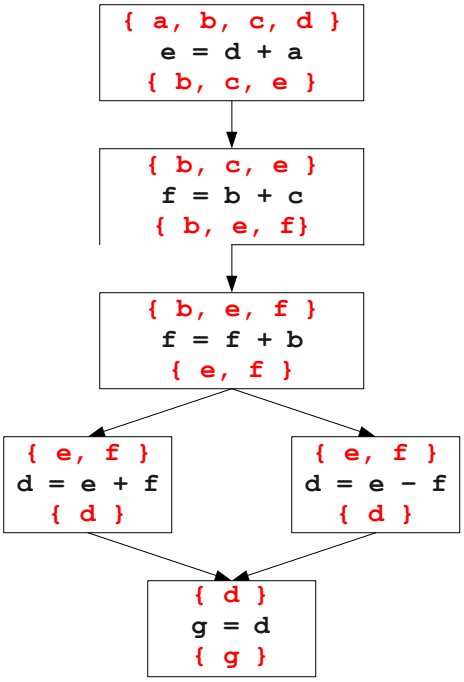

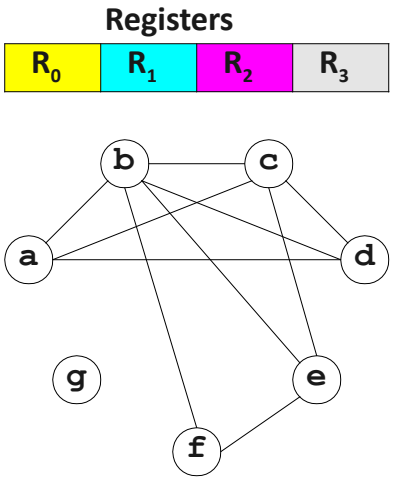

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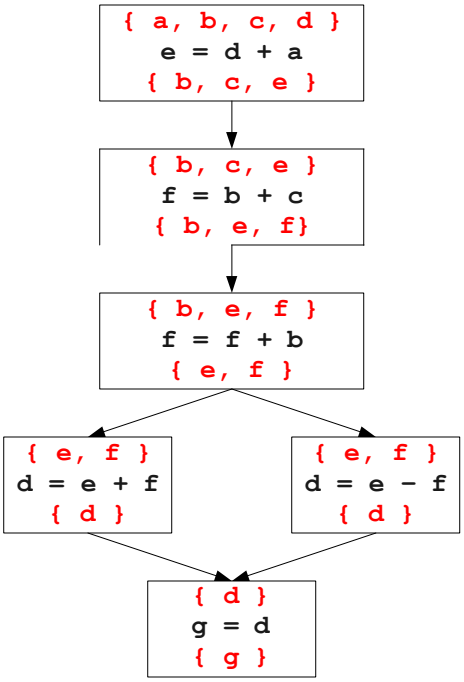
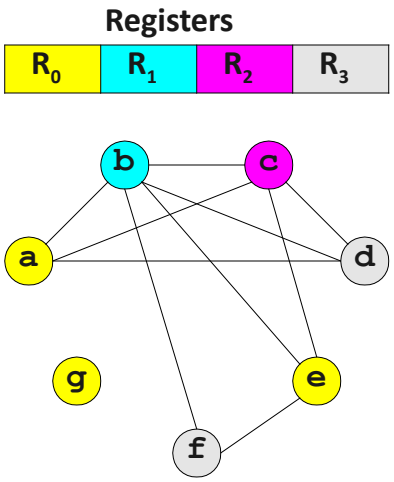
Register Interference Graph





61

The Register Interference Graph

- The **register interference graph** (RIG) of a control-flow graph is an undirected graph where
  - Each node is a variable.
  - There is an edge between two variables that are live at the same program point.
- Perform register allocation by assigning each variable a different register from all of its neighbors.

  There's just one catch...
-

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The One Catch

- This problem is equivalent to **graph-coloring**, which is **NP-hard** if there are at least three registers.
- No good polynomial-time algorithms (or even good approximations!) are known for this problem.
- We have to be content with a heuristic that is good enough for RIGs that arise in practice.

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Chaitin's Algorithm

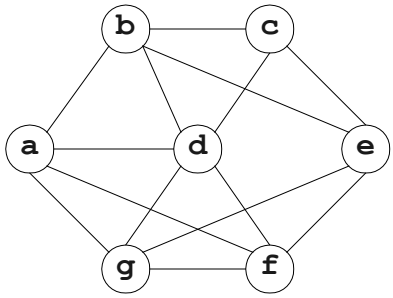
- Intuition:
  - Suppose we are trying to *k*-color a graph and find a node with fewer than *k* edges.
  - If we delete this node from the graph and color what remains, we can find a color for this node if we add it back in.

    Reason: With fewer than *k* neighbors, some color must be left over.

- Algorithm:
  - Find a node with fewer than *k* outgoing edges. Remove it
  - from the graph.
  - Recursively color the rest of the graph. Add the
  - node back in.
  - Assign it a valid color.

64

Chaitin's Algorithm



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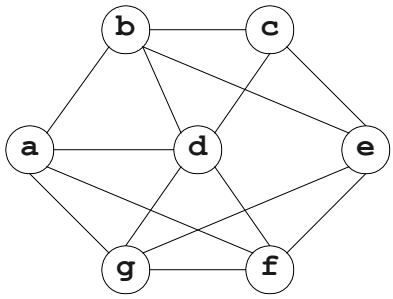
Chaitin's Algorithm

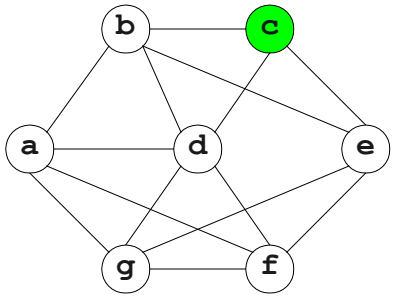


Registers

R_0	R_1	R_2	R_3

66

Chaitin's Algorithm

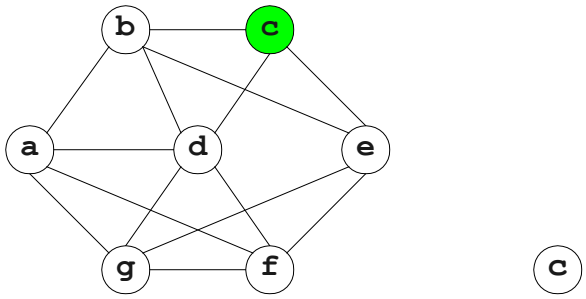


Registers

R_0	R_1	R_2	R_3

67

Chaitin's Algorithm

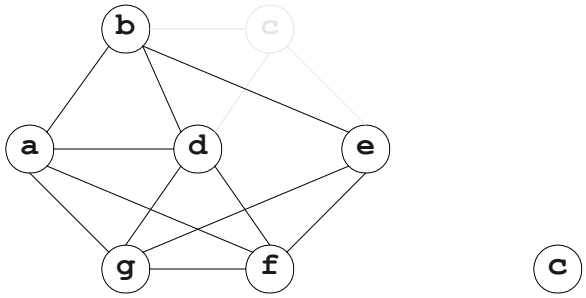


Registers

R_0	R_1	R_2	R_3

68

Chaitin's Algorithm

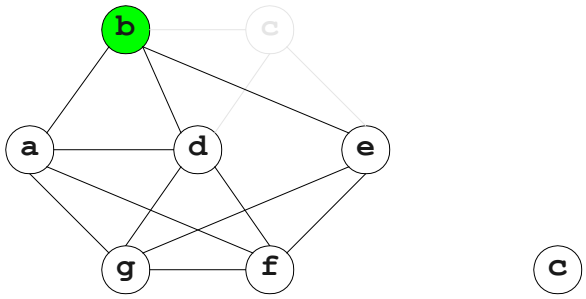


Registers

R_0	R_1	R_2	R_3

69

Chaitin's Algorithm

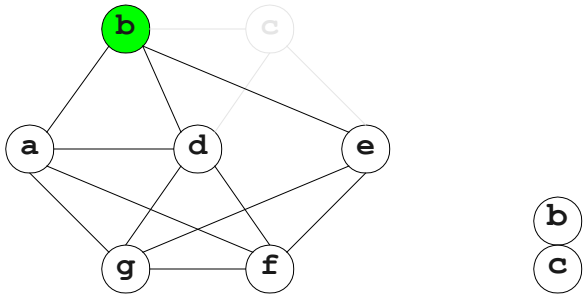


Registers

R_0	R_1	R_2	R_3

70

Chaitin's Algorithm

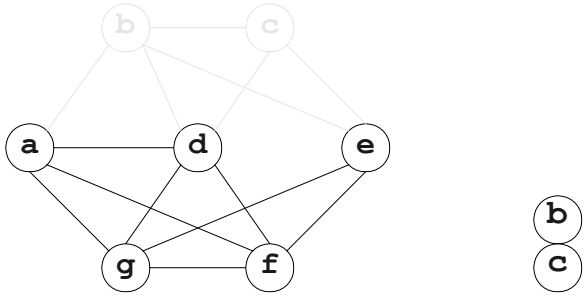


Registers

R_0	R_1	R_2	R_3

71

Chaitin's Algorithm

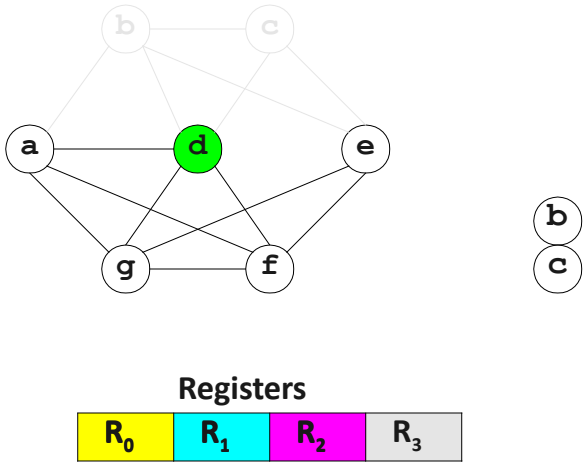


Registers

R_0	R_1	R_2	R_3

72

Chaitin's Algorithm



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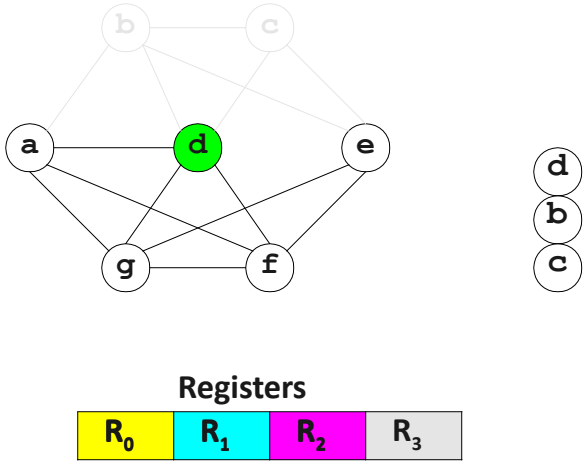
Chaitin's Algorithm



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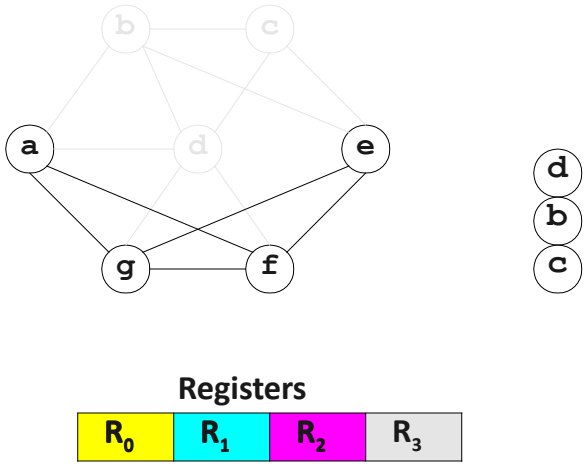
Chaitin's Algorithm



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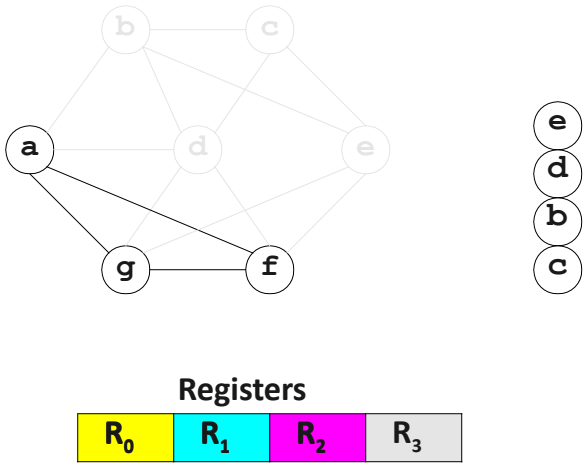
Chaitin's Algorithm



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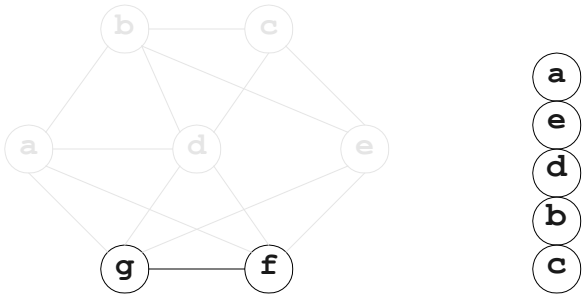
Chaitin's Algorithm

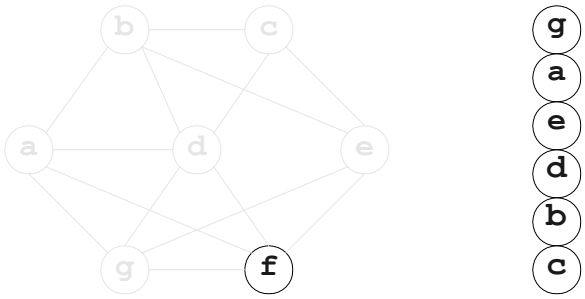


Registers

R_0	R_1	R_2	R_3

77

Chaitin's Algorithm

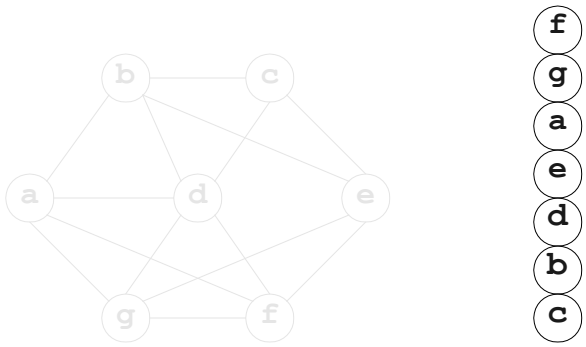


Registers

R_0	R_1	R_2	R_3

78

Chaitin's Algorithm

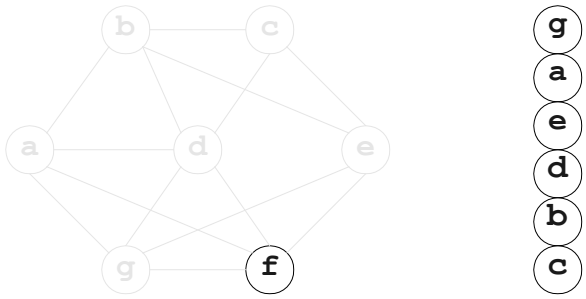


Registers

R_0	R_1	R_2	R_3

79

Chaitin's Algorithm

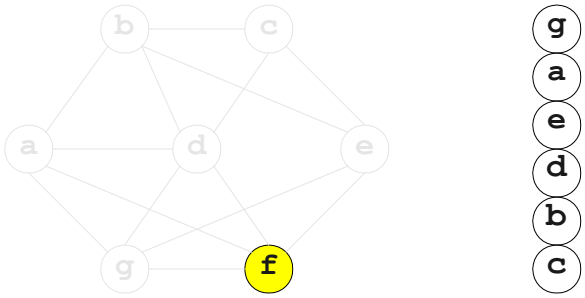


Registers

R_0	R_1	R_2	R_3

80

Chaitin's Algorithm

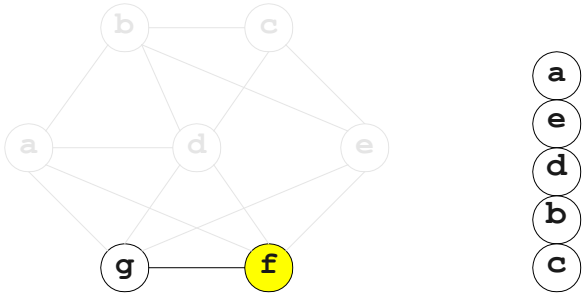


Registers

R_0	R_1	R_2	R_3

81

Chaitin's Algorithm

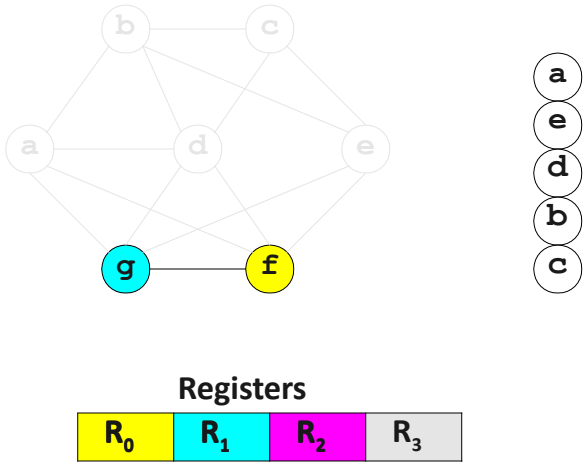


Registers

R_0	R_1	R_2	R_3

82

Chaitin's Algorithm



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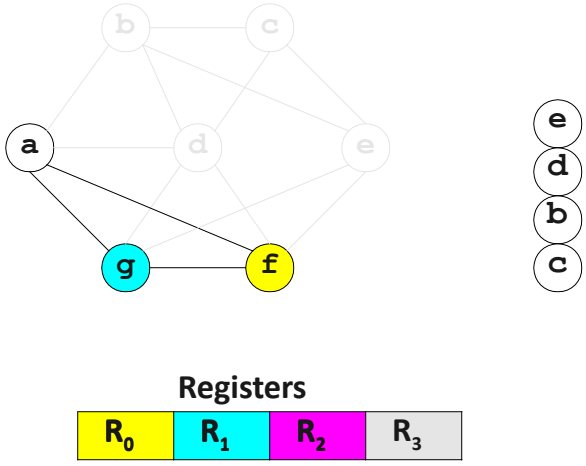
Chaitin's Algorithm



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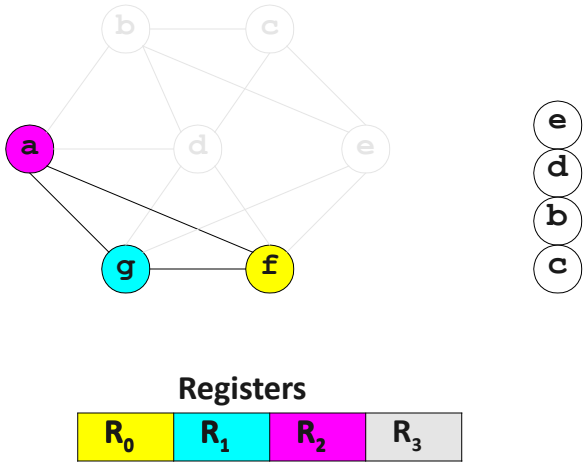
Chaitin's Algorithm



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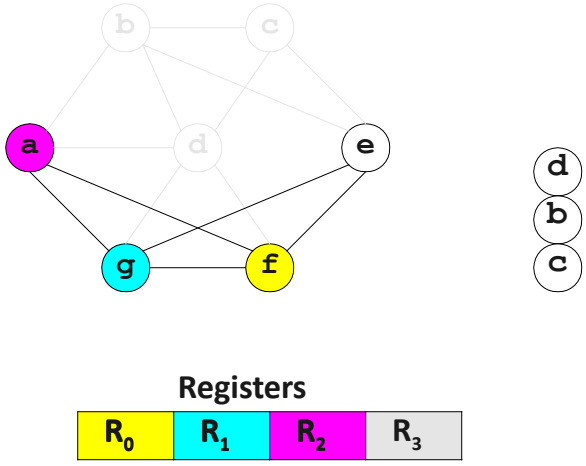
Chaitin's Algorithm



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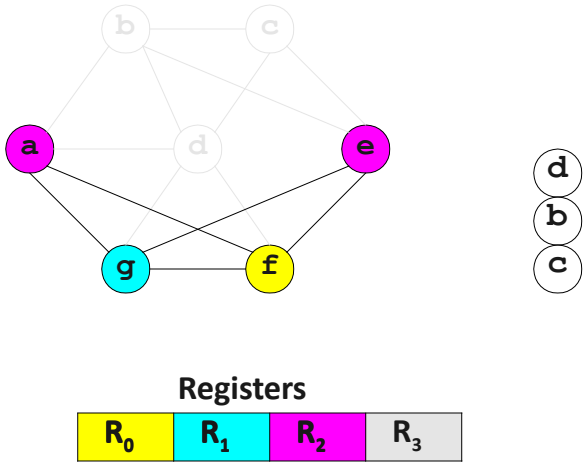
Chaitin's Algorithm



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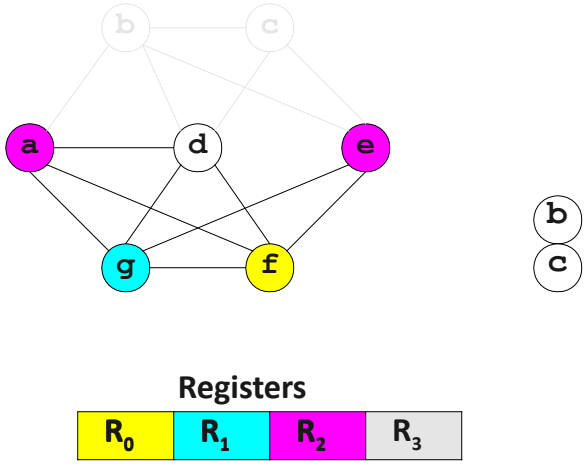

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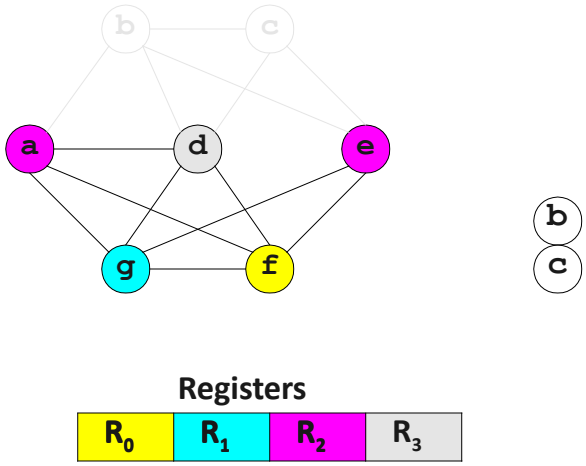

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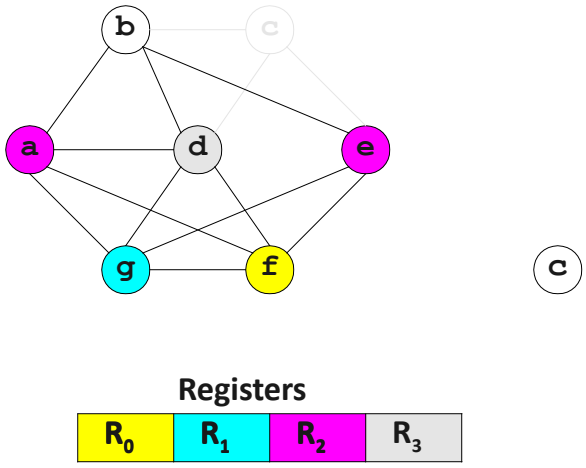

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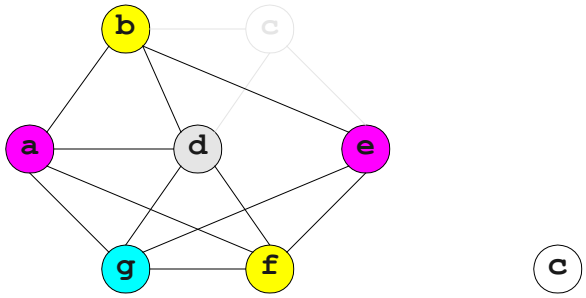
Chaitin's Algorithm

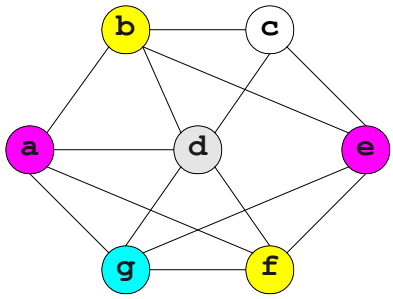


Registers

R_0	R_1	R_2	R_3

91

Chaitin's Algorithm

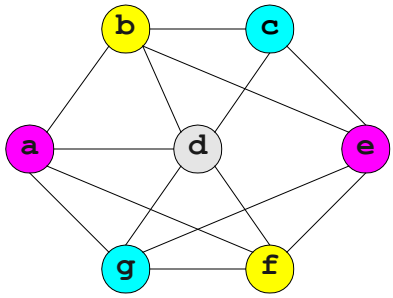


Registers

R_0	R_1	R_2	R_3

92

Chaitin's Algorithm



Registers

R_0	R_1	R_2	R_3

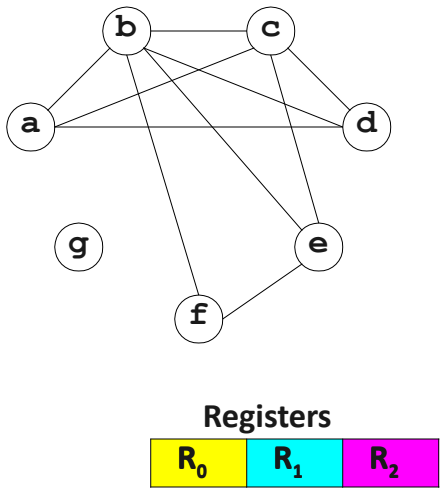
93

One Problem

- What if we can't find a node with fewer than *k* neighbors?
- Choose and remove an arbitrary node, marking it "troublesome."
  - Use heuristics to choose which one.
- When adding node back in, it may be possible to find a valid color.
- Otherwise, we have to spill that node.

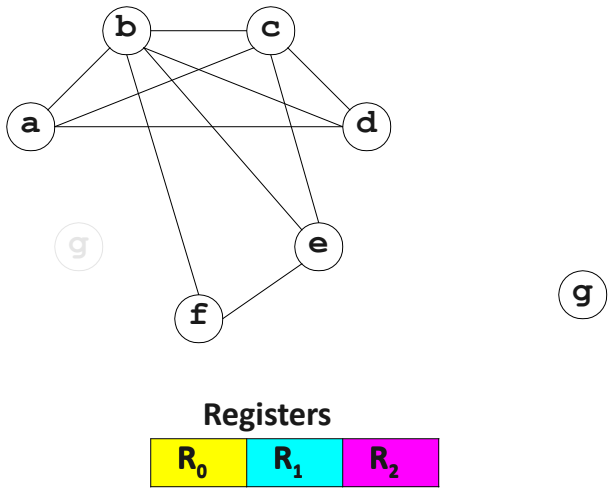
94

Chaitin's Algorithm Reloaded



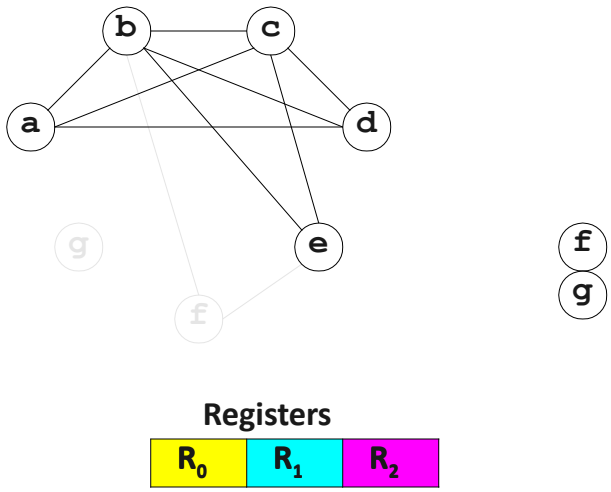
95

Chaitin's Algorithm Reloaded



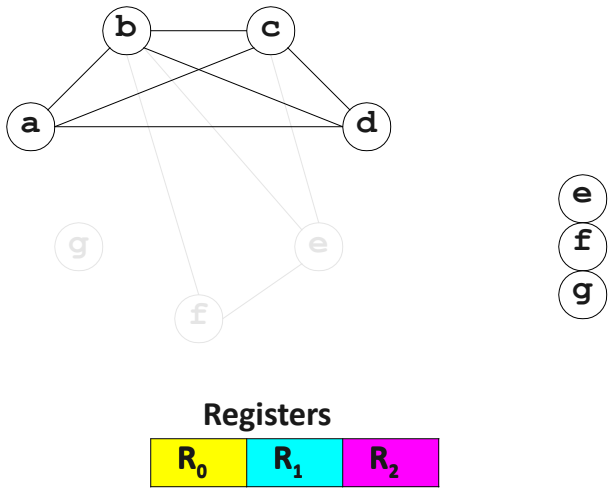
96

Chaitin's Algorithm Reloaded



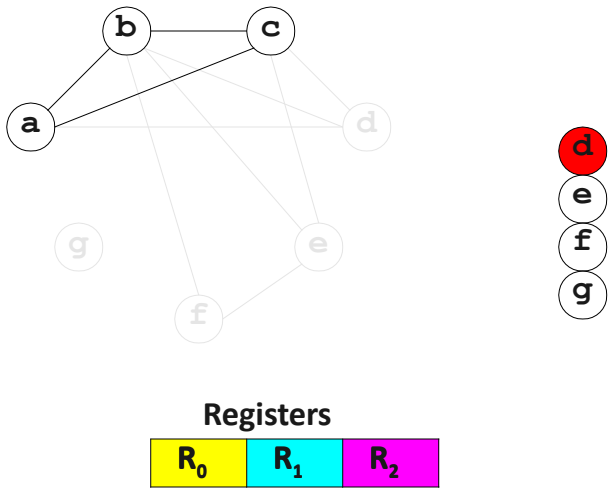
97

Chaitin's Algorithm Reloaded



98

Chaitin's Algorithm Reloaded



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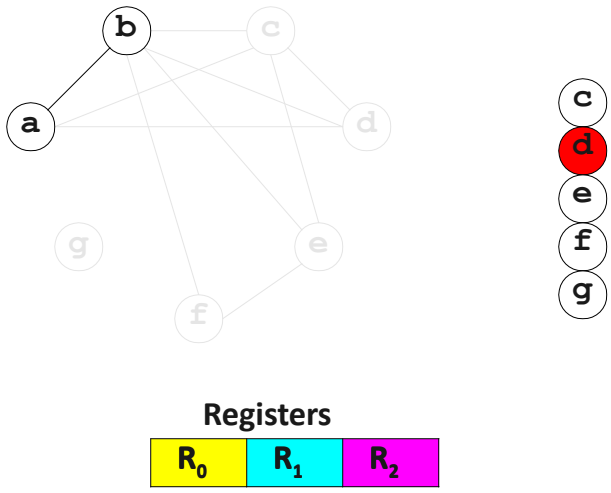

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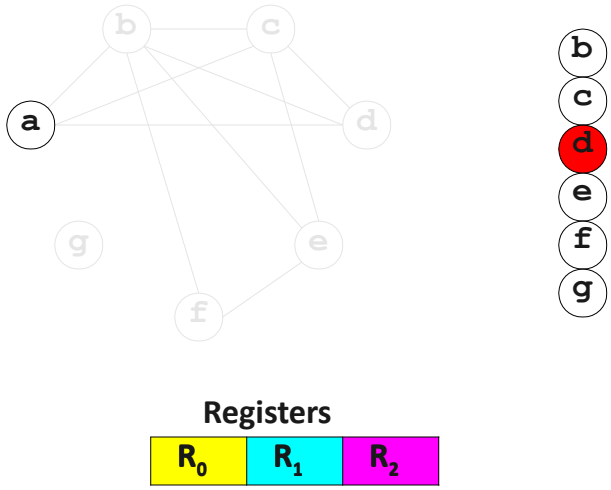

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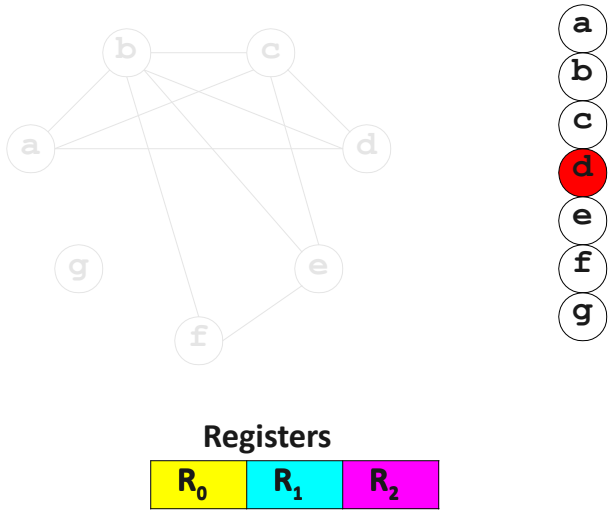

102

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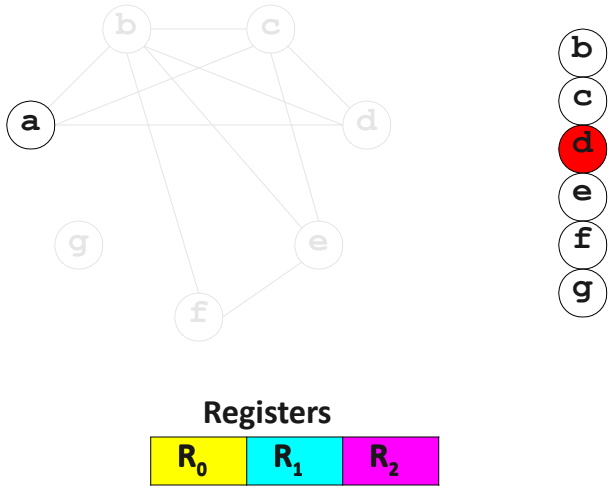

103

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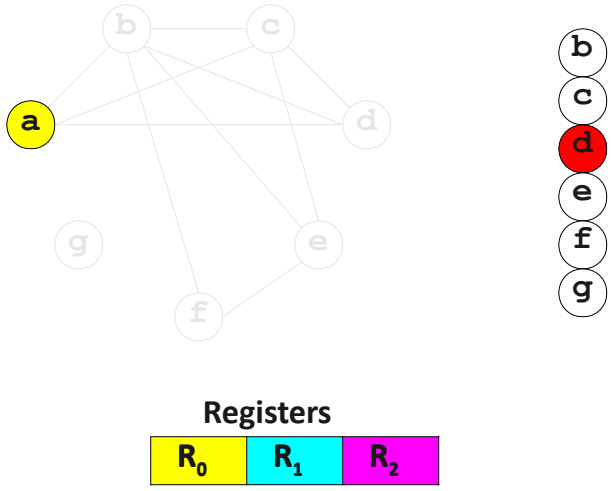

104

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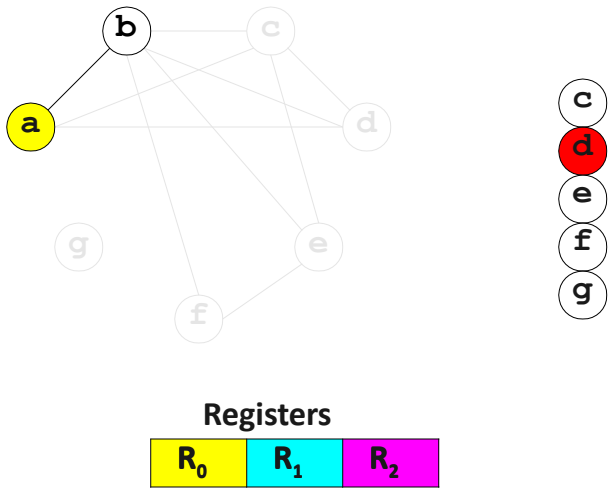

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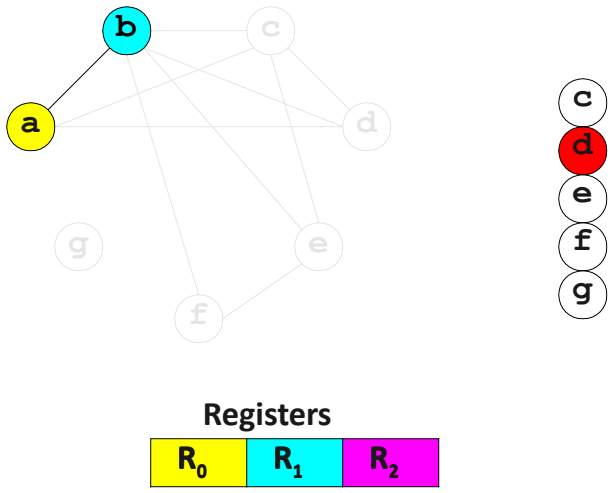

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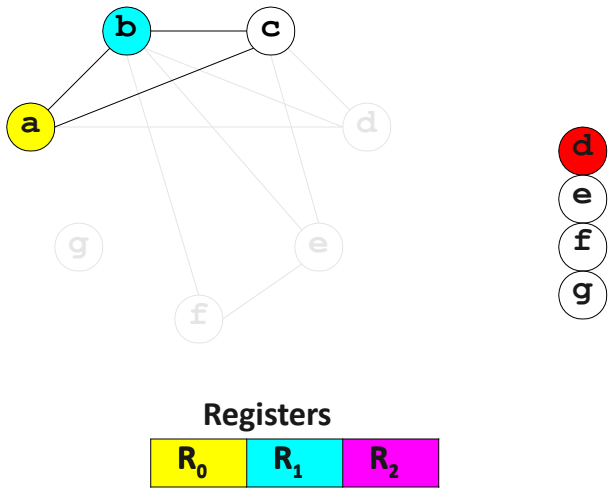

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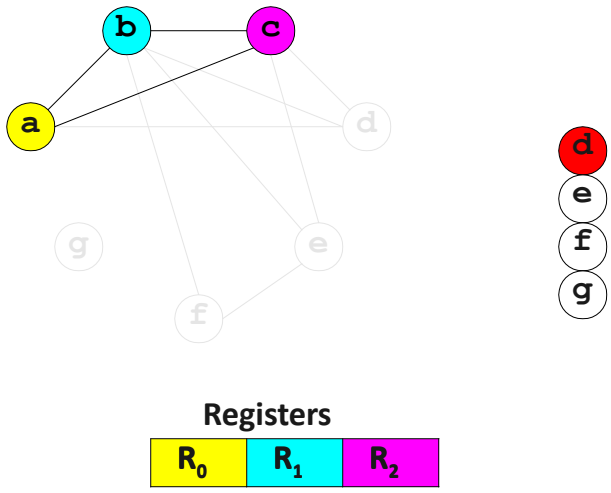

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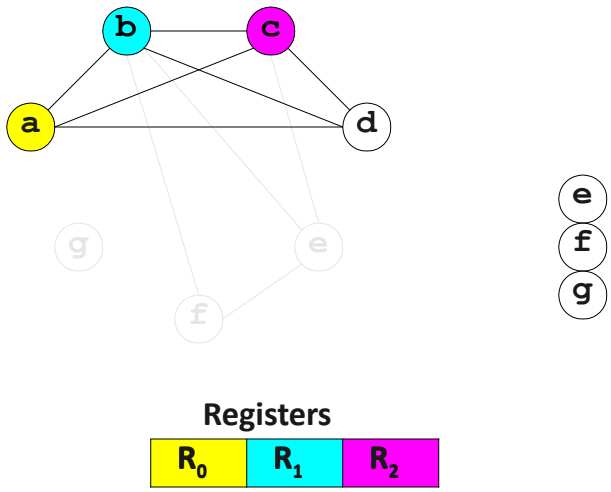

109

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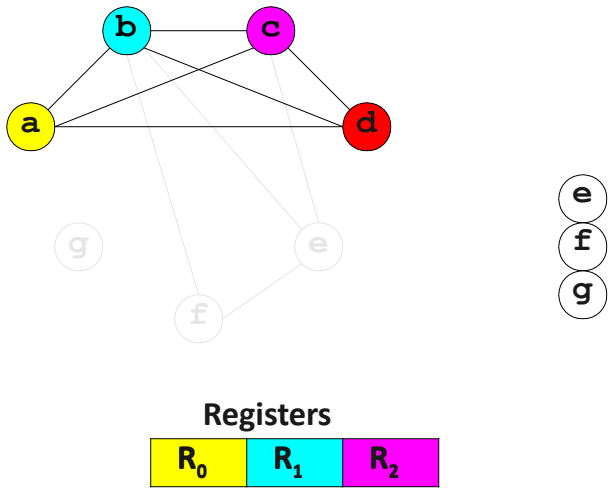

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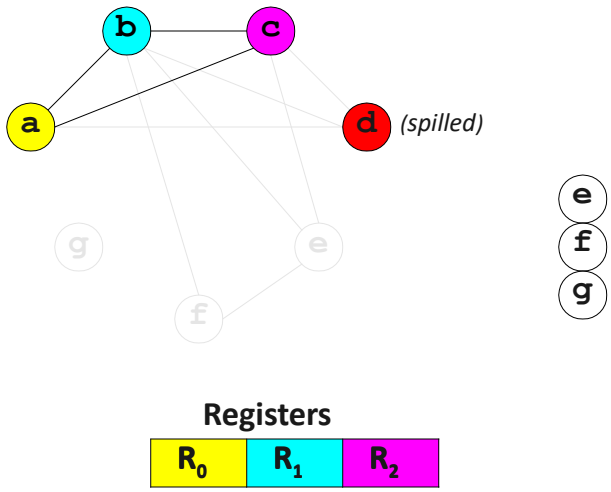

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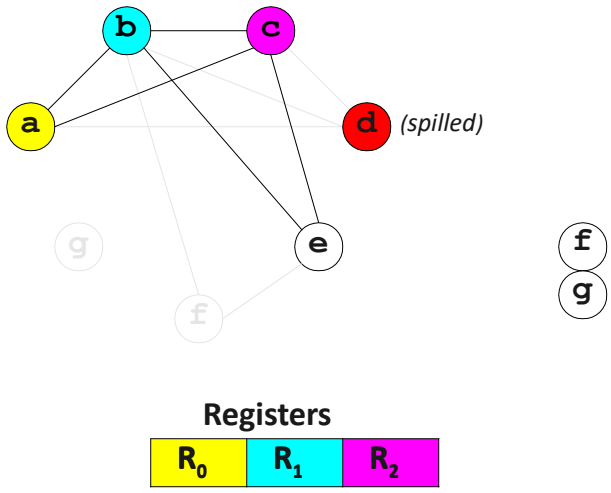

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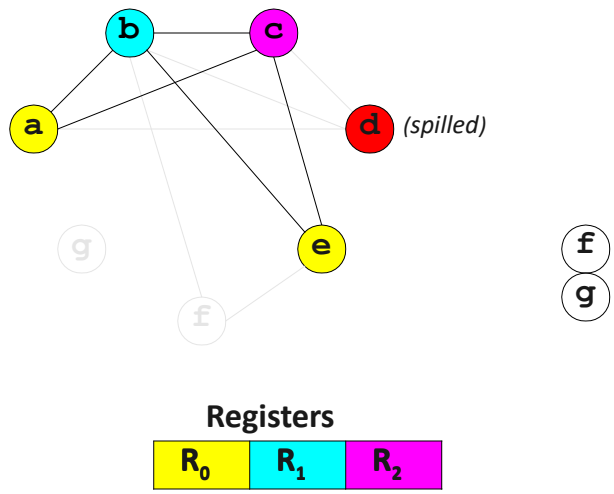

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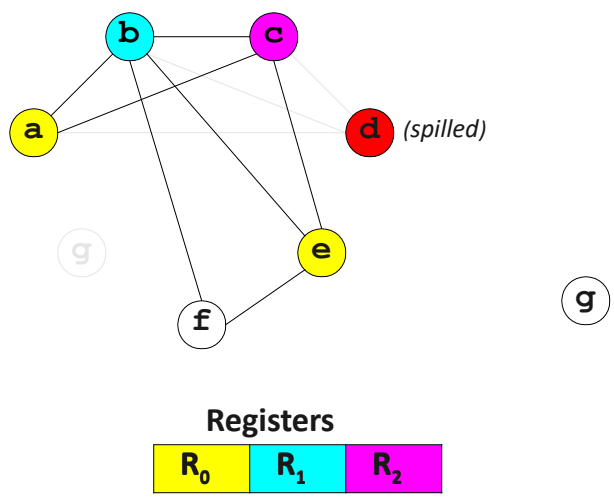

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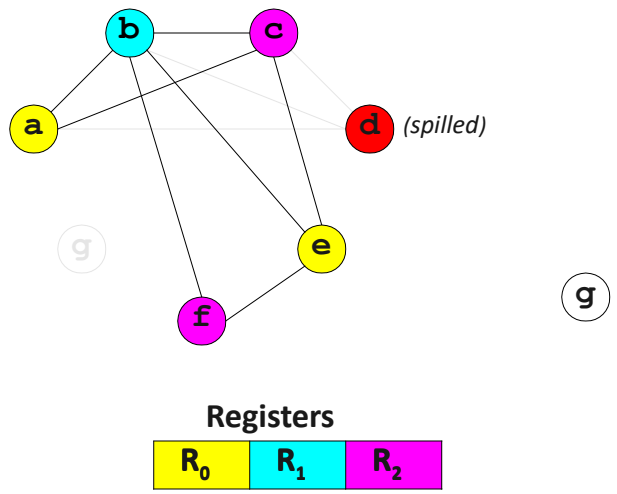

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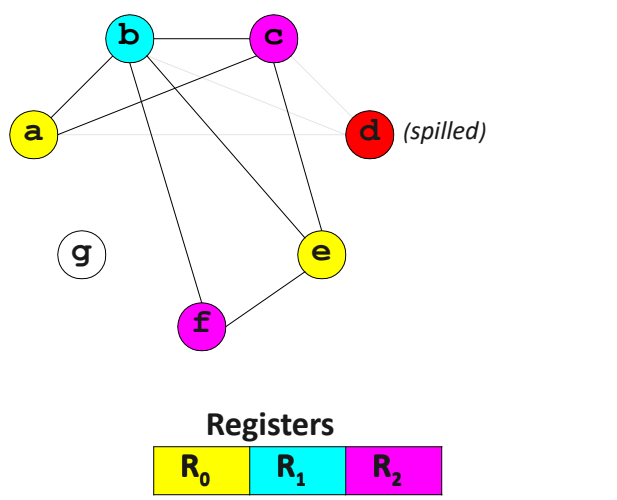

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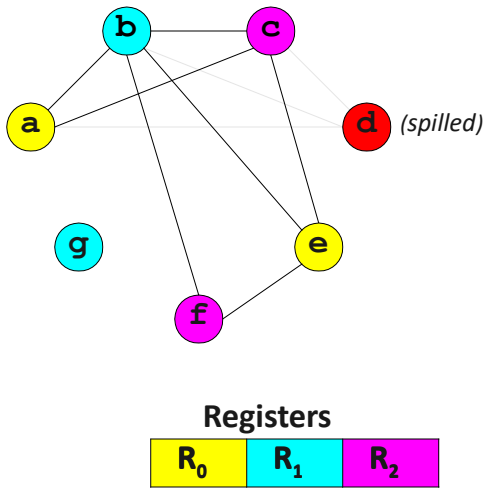
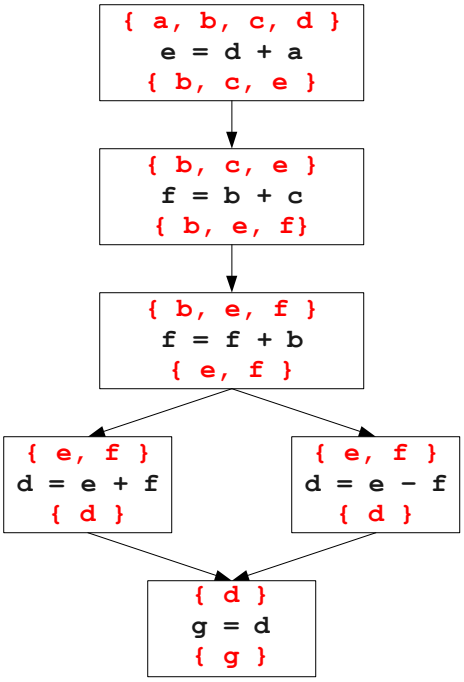
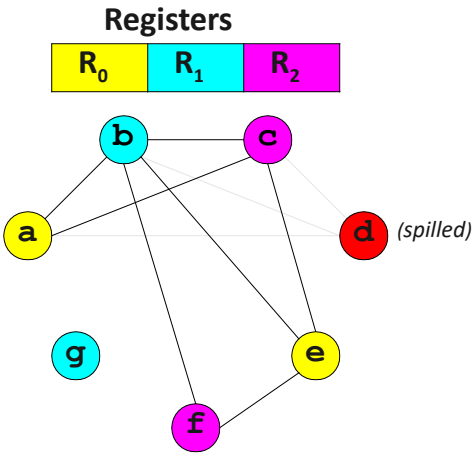


117

A Smarter Algorithm





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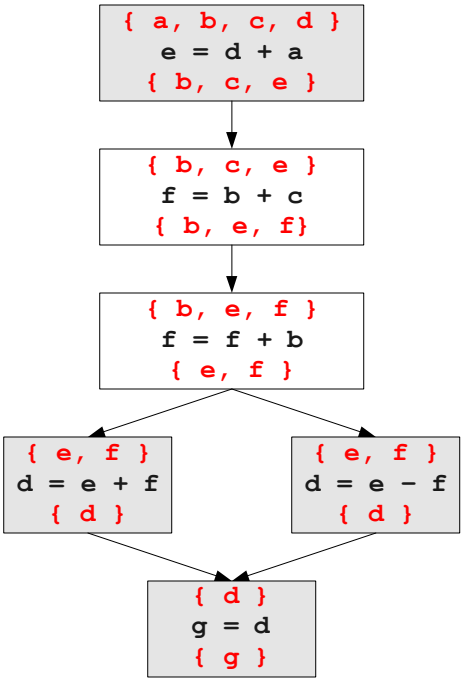
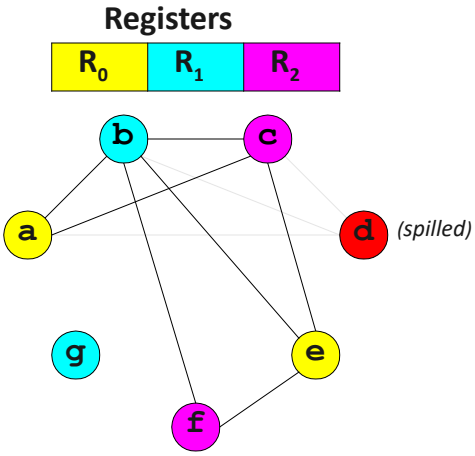
A Smarter Algorithm





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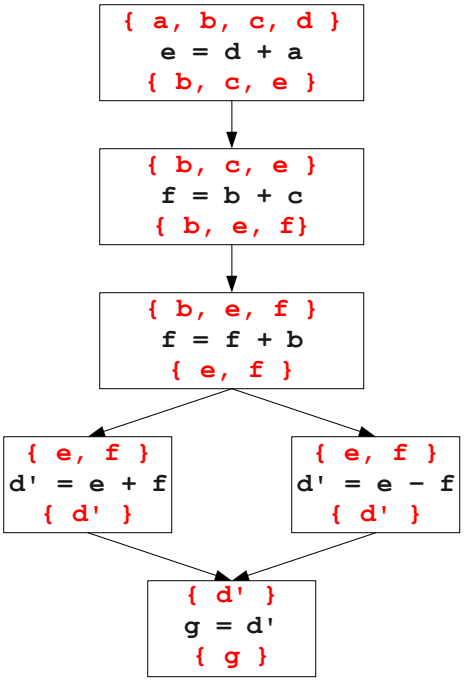
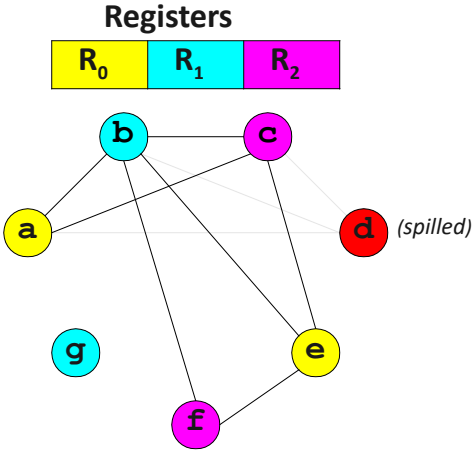
A Smarter Algorithm





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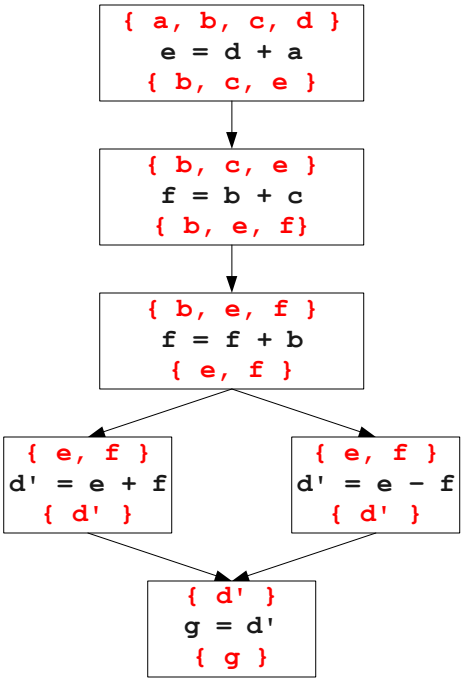
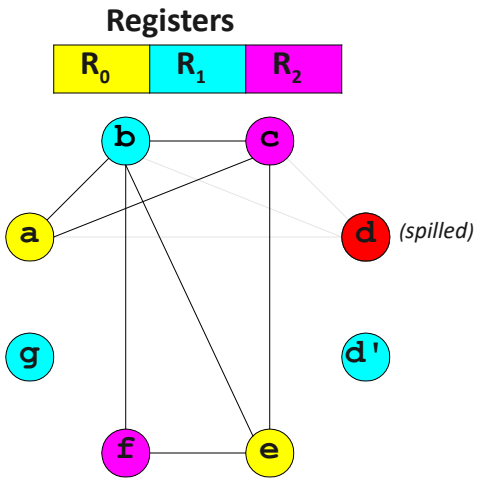
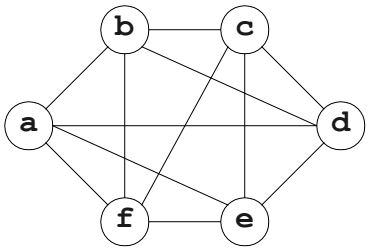
A Smarter Algorithm





121

Another Example



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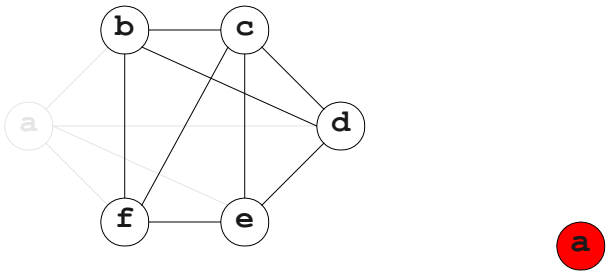
Another Example

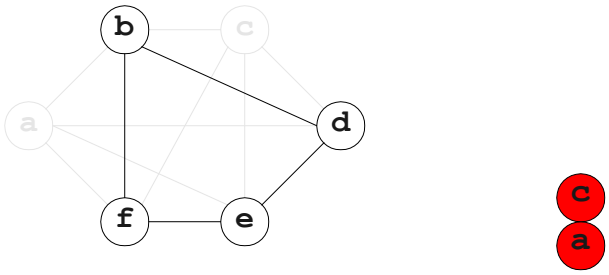


Registers

R_0	R_1	R_2

123

Another Example

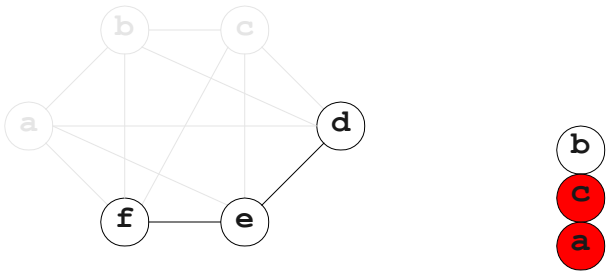


Registers

R_0	R_1	R_2

124

Another Example

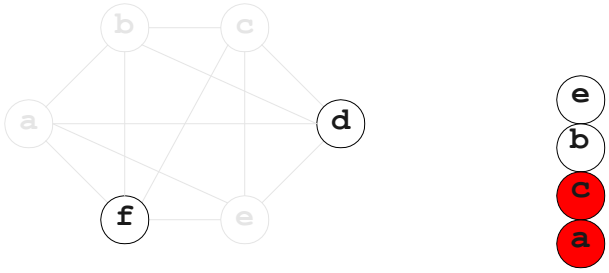


Registers

R_0	R_1	R_2

125

Another Example

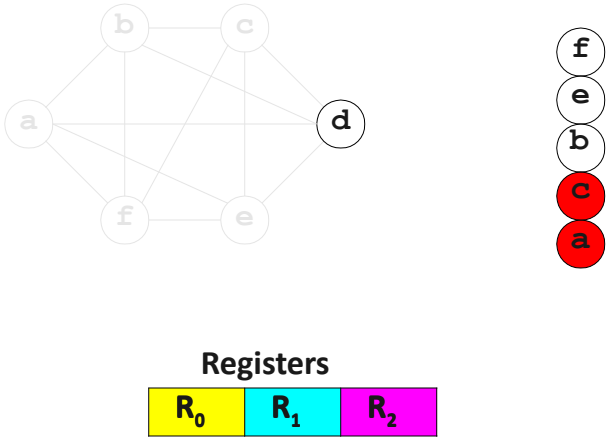


Registers

R_0	R_1	R_2

126

Another Example



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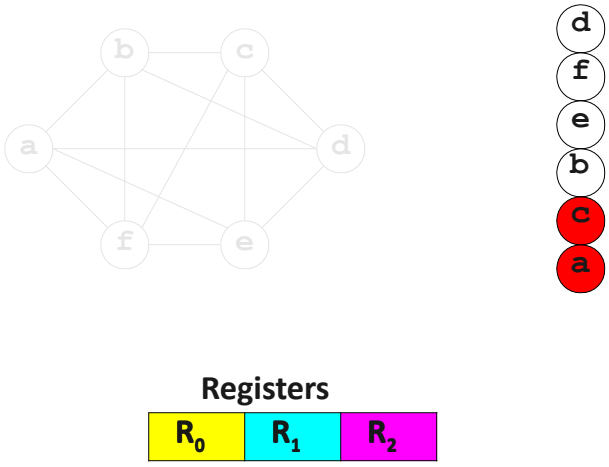
Another Example



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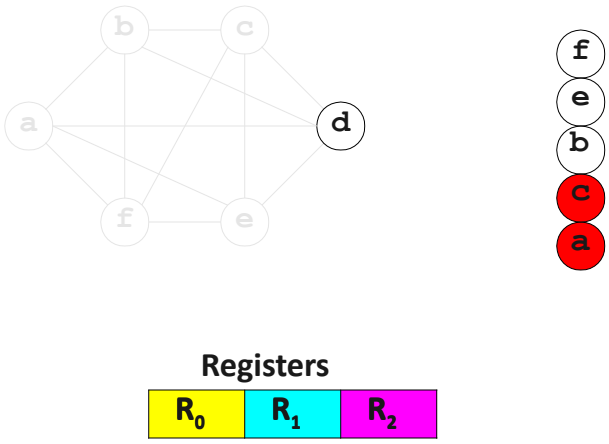
Another Example



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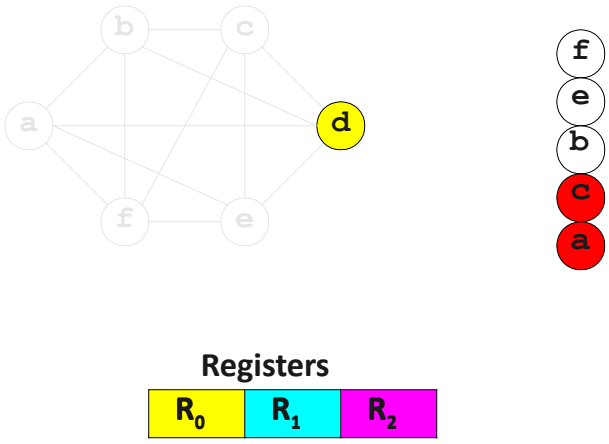
Another Example



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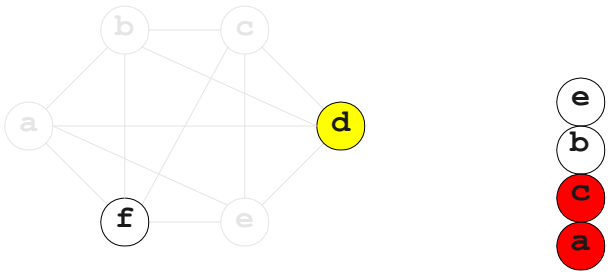
Another Example

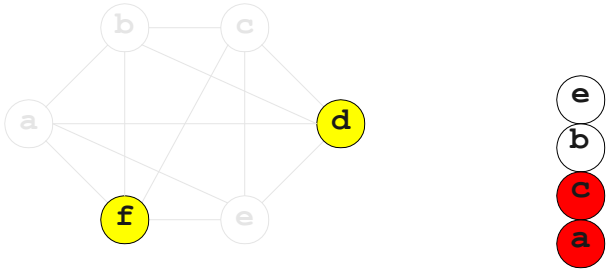


Registers

R_0	R_1	R_2

131

Another Example

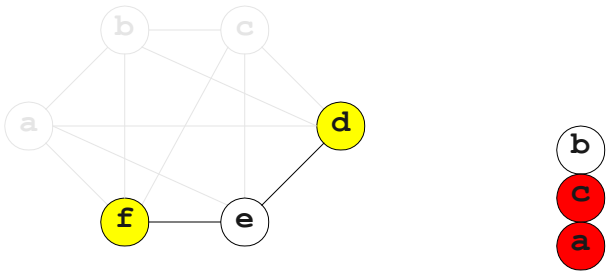


Registers

R_0	R_1	R_2

132

Another Example

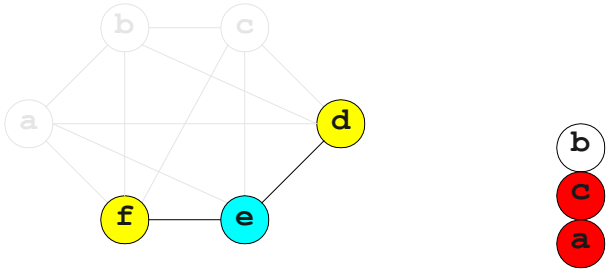


Registers

R_0	R_1	R_2

133

Another Example

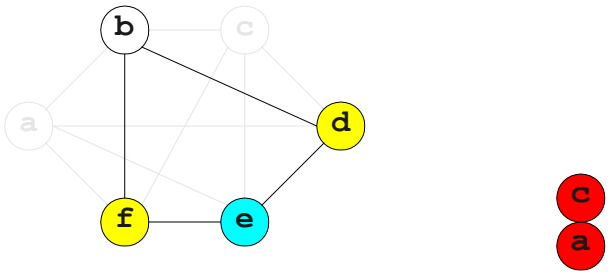


Registers

R_0	R_1	R_2

134

Another Example

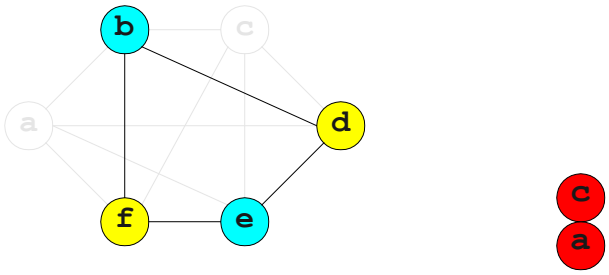


Registers

R_0	R_1	R_2

135

Another Example

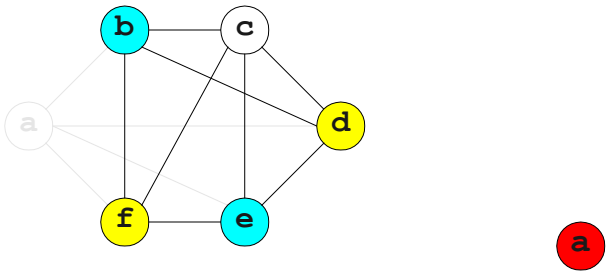


Registers

R_0	R_1	R_2

136

Another Example

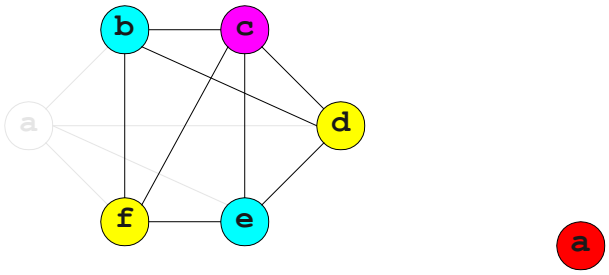


Registers

R_0	R_1	R_2

137

Another Example

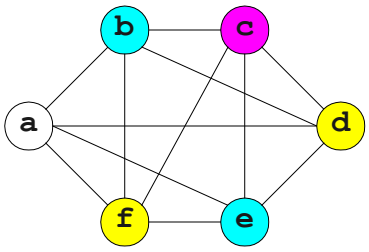


Registers

R_0	R_1	R_2

138

Another Example

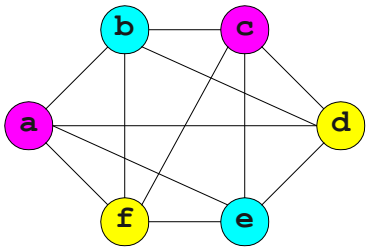


Registers

R_0	R_1	R_2

139

Another Example

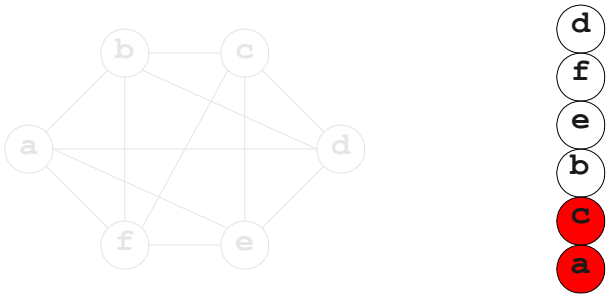


Registers

R_0	R_1	R_2

140

Another Example

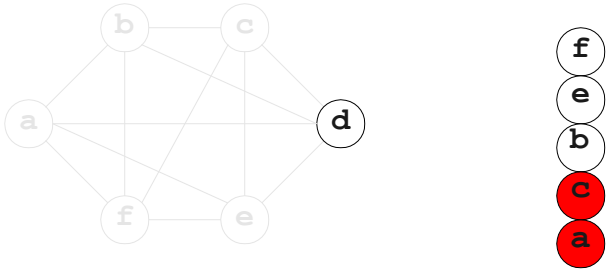


Registers

R_0	R_1	R_2

141

Another Example

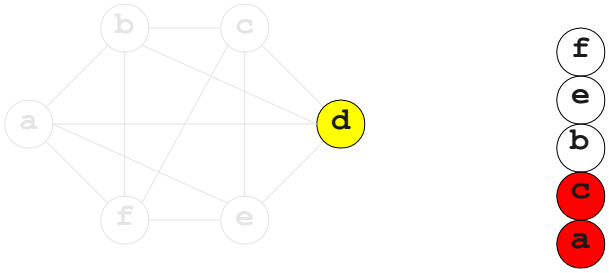


Registers

R_0	R_1	R_2

142

Another Example

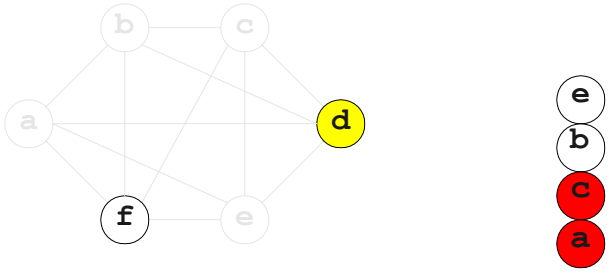


Registers

R_0	R_1	R_2

143

Another Example

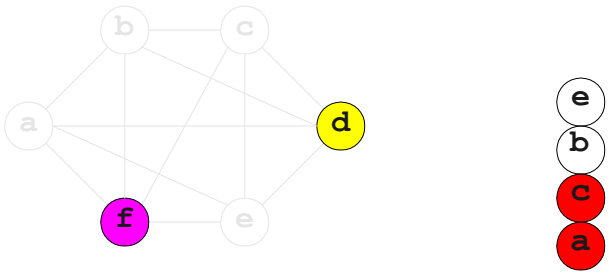


Registers

R_0	R_1	R_2

144

Another Example

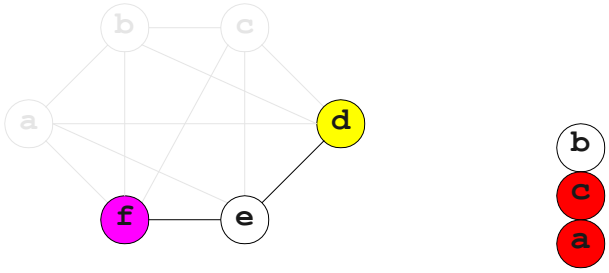


Registers

R_0	R_1	R_2

145

Another Example

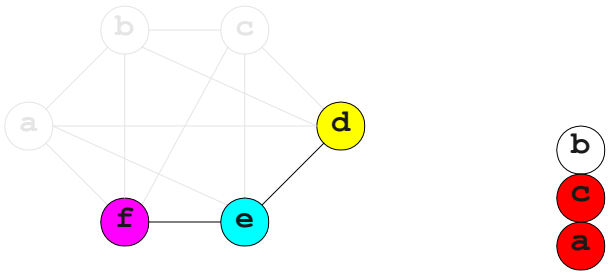


Registers

R_0	R_1	R_2

146

Another Example

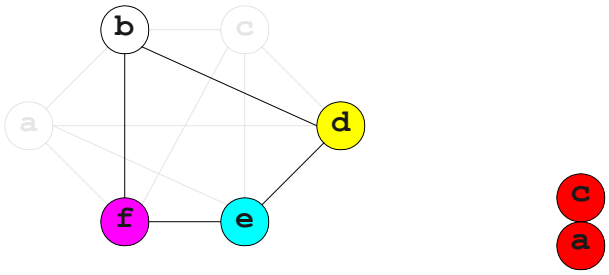


Registers

R_0	R_1	R_2

147

Another Example

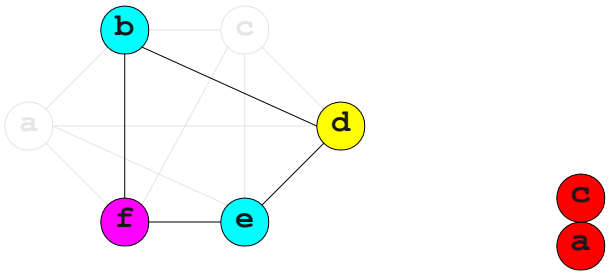


Registers

R_0	R_1	R_2

148

Another Example

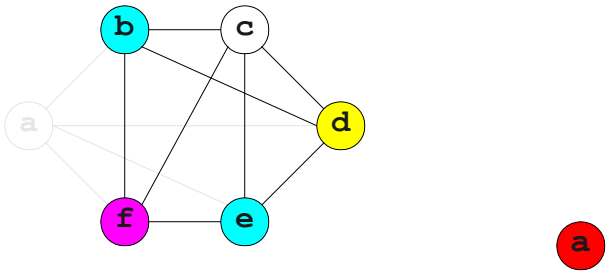


Registers

R_0	R_1	R_2

149

Another Example

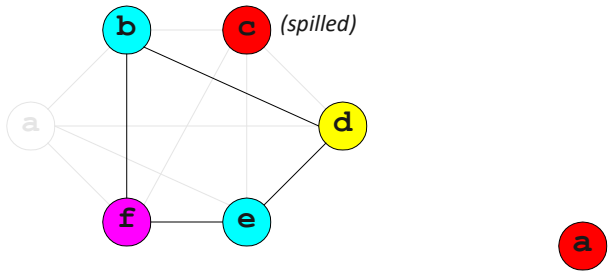


Registers

R_0	R_1	R_2

150

Another Example

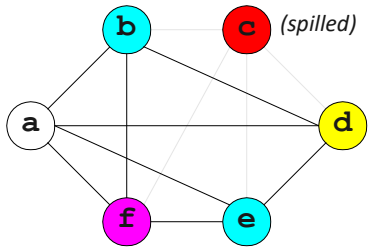


Registers

R_0	R_1	R_2

151

Another Example

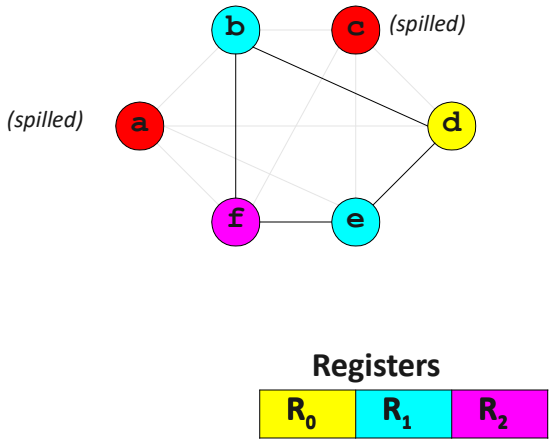


Registers

R_0	R_1	R_2

152

Another Example



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Chaitin's Algorithm

- Advantages:
  - For many control-flow graphs, finds an excellent assignment of variables to registers.
  - When distinguishing variables by use, produces a precise RIG.
  - Often used in production compilers like GCC.
- Disadvantages:
  - Core approach based on the NP-hard graph coloring problem.
  - Heuristic may produce pathologically worst-case assignments.

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Summary of Register Allocation

- Critical step in all optimizing compilers.
- The **linear scan** algorithm uses **live intervals** to greedily assign variables to registers.
  - Often used in JIT compilers due to efficiency.
- **Chaitin's algorithm** uses the **register interference graph** (based on **live ranges**) and **graph coloring** to assign registers.
  - The basis for the technique used in GCC.

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